

Generalities on algebraic groups

SGA 3, Exposé VIA

English draft

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About this draft

This is an unofficial English translation of Exposé VIA, “Generalities on algebraic groups” from *Schémas en groupes* (SGA 3). The source is the recomposition edited by P. Gille and P. Polo, PDF Exp6A-13oct24.pdf (38 pages), from <https://webusers.imj-prg.fr/~patrick.polo/SGA3/>. The seminar was directed by M. Demazure and A. Grothendieck (IHÉS, 1962–1964).

Statement numbers follow that PDF. A clear typographical slip is corrected in the English and marked with a % `typo:` comment; such slips are listed in the README. A slip that would change the mathematics is left as printed and marked % `typo?:`. Scholarly proofreading remains outstanding.

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Exposé VIA. Generalities on algebraic groups

by P. Gabriel

⁽¹⁾Throughout this chapter, A will denote an artinian local ring with residue field k . A group scheme G over $\mathrm{Spec} A$ will simply be called an A -group. This A -group is said to be locally of finite type if the underlying scheme is locally of finite type over A ; it is said to be algebraic if the underlying scheme is of finite type over A .

0. Preliminary remarks

0.1. — First consider a group scheme G over any scheme S . We call the structure morphism $\mu : G \times_S G \rightarrow G$ multiplication, and the morphism $c : G \rightarrow G$ defined by the equalities $c(T)(x) = x^{-1}$ inversion (T being a scheme over S and x an element of $G(T)$). If U and V are subsets of the underlying set of G , we denote by $U \cdot V$ the image under the multiplication morphism of the subset of $G \times_S G$ consisting of the points whose first projection belongs to U and second to V . Similarly, the notations U^{-1} and $c(U)$ are equivalent.

Let pr_1 be the projection of $G \times_S G$ onto the first factor, and $\sigma : G \times_S G \rightarrow G \times_S G$ the morphism with components pr_1 and μ . For every S -scheme T , $\sigma(T)$ is the map $(x, y) \mapsto (x, xy)$; it follows that σ is an automorphism. The composite of this automorphism with the projection pr_2 of $G \times_S G$ onto the second factor is the multiplication morphism. When G is flat over S , pr_2 , and hence μ , are flat morphisms; when G is smooth over S , pr_2 , and hence μ , are smooth morphisms, etc.

0.2. — From now on we suppose that S is the spectrum of an artinian local ring A with residue field k . We denote by $(\mathrm{Sch}/k)_{\mathrm{red}}$ the category of reduced schemes over k . For every scheme X over A , the reduced scheme X_{red} is an object of $(\mathrm{Sch}/k)_{\mathrm{red}}$, and the functor $X \mapsto X_{\mathrm{red}}$ is right adjoint to the inclusion of $(\mathrm{Sch}/k)_{\mathrm{red}}$ into (Sch/A) . It follows that, for every A -group G , G_{red} is a group in the category $(\mathrm{Sch}/k)_{\mathrm{red}}$,⁽²⁾ that is, for every reduced k -scheme T , $G_{\mathrm{red}}(T)$ has a group structure functorial in T . One must beware that G_{red} is not necessarily a k -group, for “multiplication” is only a morphism from $(G_{\mathrm{red}} \times_k G_{\mathrm{red}})_{\mathrm{red}}$ to G_{red} .⁽³⁾

⁽⁴⁾However, if k is a perfect field, the inclusion of $(\mathrm{Sch}/k)_{\mathrm{red}}$ into (Sch/k) commutes with products, so that groups in the category $(\mathrm{Sch}/k)_{\mathrm{red}}$ are identified with group schemes over k whose underlying scheme is reduced. In this case, if G is a k -group, G_{red} is a subgroup scheme of G ; this subgroup is not in general invariant in G .

For example, if k is a field of characteristic 3, the constant group $(\mathbf{Z}/2\mathbf{Z})_k$ acts nontrivially on the diagonalizable group $D_k(\mathbf{Z}/3\mathbf{Z})$ (cf. Exp. I, 4.1 and 4.4); if G denotes the semidirect product of $D_k(\mathbf{Z}/3\mathbf{Z})$ by $(\mathbf{Z}/2\mathbf{Z})_k$ defined by this action, G_{red} is identified with $(\mathbf{Z}/2\mathbf{Z})_k$ and is not invariant in G .⁽⁵⁾

⁽¹⁾**N.D.E.** Version of 13 October 2024.

⁽²⁾**N.D.E.** The following sentences have been added.

⁽³⁾**N.D.E.** For examples of group schemes G over an imperfect field k such that G_{red} is not a k -group scheme, see 1.3.2 below.

⁽⁴⁾**N.D.E.** What follows in 0.2, as well as 0.3, has been slightly modified.

⁽⁵⁾**N.D.E.** For an analogous example with G connected, one can consider, for $\mathrm{char}(k) = p > 0$, the semidirect product of $\alpha_{p,k}$ and $\mathbf{G}_{m,k}$.

Let k be any field, $k^{p^{-\infty}}$ its perfect closure, and H a group in the category $(\text{Sch}/k)_{\text{red}}$. Then $(H \otimes_k k^{p^{-\infty}})_{\text{red}}$ is a group scheme over the perfect field $k^{p^{-\infty}}$. Since $H \otimes_k k^{p^{-\infty}}$ and $(H \otimes_k k^{p^{-\infty}})_{\text{red}}$ have the same underlying topological space, one sees that groups in $(\text{Sch}/k)_{\text{red}}$ share with k -groups certain topological properties invariant under extension of the base field: for example, it will follow from 0.3 and the remarks just made that every group in $(\text{Sch}/k)_{\text{red}}$ is separated.

We shall encounter groups in $(\text{Sch}/k)_{\text{red}}$ below whenever we deal with a nonempty locally closed subset U of an A -group G such that $U \cdot U = U$ and $U^{-1} = U$: indeed, the reduced subscheme of G defined by U is a group in $(\text{Sch}/k)_{\text{red}}$.

0.3. — An A -group G is always separated, for the unit section $e : \text{Spec } A \rightarrow G$ is a closed immersion. Indeed, let x be the unique point of $\text{Spec } A$ and η the structure morphism $G \rightarrow \text{Spec } A$. Since $\eta \circ e = \text{id}_{\text{Spec } A}$, for every affine open subset $U = \text{Spec } B$ of G containing $e(x)$ the morphism $B \rightarrow A$ has a section, hence is surjective. It follows that e is a closed immersion.⁽⁶⁾ Now the diagonal of $G \times_A G$ is identified with the functor from $(\text{Sch}/A)^\circ$ to (Ens) assigning to every scheme S over A the inverse image of the neutral element of $G(S)$ under the map $\varphi(S) : (x, y) \mapsto x \cdot y^{-1}$ from $G(S) \times G(S)$ to $G(S)$. One consequently has the cartesian square below, so the diagonal morphism, obtained from a closed immersion by base change, is itself a closed immersion:

$$\begin{array}{ccc} G \times_S G & \xrightarrow{\varphi} & G \\ \uparrow \text{diagonal morphism} & & \uparrow \text{unit section} \\ G & \longrightarrow & \text{Spec } A. \end{array}$$

0.4. — ⁽⁷⁾Let G be an A -scheme. We shall say that a point g of G is strictly rational over A if there exists an A -morphism $s : \text{Spec } A \rightarrow G$ sending the only point of $\text{Spec } A$ to g , i.e. if the morphism $A \rightarrow \mathcal{O}_{G,g}$ admits a retraction. Note that one then has $\kappa(g) = k$, and therefore g is a closed point of G (if B is the ring of an affine open neighborhood of g and $\mathfrak{p}$ the prime ideal of B corresponding to g , then $B/\mathfrak{p} \subset k$ is a finite A -algebra, hence an integral artinian ring, hence a field).

Henceforth suppose G an A -group; then such an $s : \text{Spec } A \rightarrow G$ defines an automorphism r_s of the scheme G over A , called right translation by s : for every morphism $\pi : S \rightarrow \text{Spec } A$, $r_s(\pi)$ is the automorphism of $G(S)$ defined by $x \mapsto x \cdot G(\pi)(s)$ for every $x \in G(S)$. Similarly, ℓ_s denotes left translation by s , that is, the automorphism of G defined by $\ell_s(\pi)(x) = G(\pi)(s) \cdot x$ for every $x \in G(S)$.

Since $G \otimes_A k$ and G have the same underlying topological space $\underline{G}$, $G \otimes_A k$ is a k -group, and $s \otimes_A k$ depends only on g and not on s , one sees that the automorphisms of $\underline{G}$ induced by r_s and ℓ_s (or by $r_{s \otimes k}$ and $\ell_{s \otimes k}$) depend only on g and not on s ; when P is a subset of $\underline{G}$, we can therefore write $r_g(P)$ or $P \cdot g$ (resp. $\ell_g(P)$ or $g \cdot P$) instead of $r_s(P)$ (resp. $\ell_s(P)$), consistently with 0.1.

Remark 0.4.1. If g is a strictly rational point of G and $A \rightarrow A'$ is a morphism of artinian local rings, then $G' = G \otimes_A A'$ has a unique point g' above g , and g' is strictly rational over A' ; moreover, if P' denotes the inverse image of P in G' , then $P' \cdot g'$ is the inverse image of $P \cdot g$ (cf. EGA I, 3.4.8).

⁽⁶⁾**N.D.E.** This argument holds for any zero-dimensional local ring A , and shows that if S is a discrete scheme, every S -group is separated, cf. VIB, 5.2; on the other hand (cf. VIB, 5.6.4), if S contains a nonisolated closed point s , the S -scheme G obtained by gluing two copies of S along the open subset $S - \{s\}$ is not separated over S , but has an S -group structure.

⁽⁷⁾**N.D.E.** The original has been expanded in what follows; in particular, Remark 0.4.1 has been added.

Proposition 0.5. ⁽⁸⁾ *Let G be an A -group and U, V two dense open subsets of G . Then $U \cdot V$ (i.e. the image of $U \times_A V$ under multiplication) equals the whole underlying space of G .*

Indeed, since G and $G \otimes_A k$ have the same underlying space, one can suppose, replacing A by k and G by $G \otimes_A k$ if necessary, that $A = k$. Let $g \in G$. Put $K = \kappa(g)$; then left translation ℓ_g is an automorphism of G_K . Since the projection $G_K \rightarrow G$ is open, U_K and V_K are dense open subsets of G_K , as is the image of V_K under λ_g . There therefore exists $v \in V_K$ such that $u = \ell_g(v)$ belongs to U_K . Let L be an extension of K containing $\kappa(v)$, and hence $\kappa(u)$, and let g_L and v_L be the L -points of G_L obtained from g and v . Then $g_L \cdot v_L = u'$ is a point of G_L above u , and therefore $g_L = u' \cdot v_L^{-1}$ lies above $U \cdot V$, whence $g \in U \cdot V$, which proves the proposition.

Corollary 0.5.1. ⁽⁹⁾ *If G is an irreducible A -group, G is quasi-compact.*

Indeed, let U be a nonempty affine open subset of G ; then U is dense in G , so, by 0.5, the morphism $\mu : U \times_A U \rightarrow G$ is surjective, hence G is quasi-compact since $U \times_A U$ is.

Corollary 0.5.2. ⁽¹⁰⁾ *Let G be an A -group scheme and H a subgroup A -scheme of G . Then H is closed.*

PROOF. Let k' be the perfect closure of the residue field k of A . Since the underlying topological spaces of G and H are unchanged by the base change $A \rightarrow k \rightarrow k'$, one can suppose $A = k$ a perfect field. One can then suppose G and H reduced, hence geometrically reduced.

Let $\overline{H}$ be the closure of H ; then $\mu^{-1}(\overline{H})$ is a closed subset of $G \times G$ containing $H \times H$. Since the morphism $H \rightarrow \text{Spec } k$ (resp. $\overline{H} \rightarrow \text{Spec } k$) is universally open, and H is dense in $\overline{H}$, $H \times H$ is dense in $H \times \overline{H}$, and $H \times \overline{H}$ is dense in $\overline{H} \times \overline{H}$, so $H \times H$ is dense in $\overline{H} \times \overline{H}$. Thus $\mu(\overline{H} \times \overline{H}) \subset \overline{H}$, and hence, since $\overline{H} \times \overline{H}$ is reduced, μ induces a morphism $\mu' : \overline{H} \times \overline{H} \rightarrow \overline{H}$.

Let, then, $g \in \overline{H}$ and put $K = \kappa(g)$. Since the projection $\overline{H}_K \rightarrow \overline{H}$ is open, H_K and $\ell_g(H_K)$ are two dense open subsets of $\overline{H}_K$, so there exist $u, v \in H_K$ such that $\ell(g)(v) = u$. One concludes, as in the proof of 0.5, that g belongs to $H \cdot H = H$, whence $\overline{H} = H$. $\square$

1. Local properties of an A -group locally of finite type

We shall first see that, if G is locally of finite type and flat over A , one can “make every closed point of G strictly rational” by a finite flat extension of the base.

1.1. — Unless expressly stated otherwise, we shall henceforth suppose G an A -group locally of finite type. When A is a field k , we shall obtain in Exposé VIIIB very precise results on the local rings of G .⁽¹¹⁾ Here we content ourselves with some elementary results:

Proposition 1.1.1. *Let x be a point of an A -group G locally of finite type and flat over A . Then the local ring $\mathcal{O}_{G,x}$ is Cohen–Macaulay, and there exists a system $a_1, \dots, a_n$ of parameters of $\mathcal{O}_{G,x}$ such that $\mathcal{O}_{G,x}/(a_1, \dots, a_n)$ is a finite flat (hence finite free) A -module.*

We first suppose A equal to its residue field k ; it then suffices to prove that $\mathcal{O}_{G,x}$ is Cohen–Macaulay, and one can restrict to the case where x is a closed point (cf. EGA 0_{IV}, 16.5.13). By Lemma 1.1.2 below, G contains a closed point y such that $\mathcal{O}_{G,y}$ is Cohen–Macaulay. By

⁽⁸⁾**N.D.E.** The original stated this result under the hypothesis that G is locally of finite type over A . Since it is useful to have it in the general case, and the proof is essentially the same, the result has been stated and proved in the general case. This will be used several times below.

⁽⁹⁾**N.D.E.** This corollary has been inserted here, cf. 2.4 and 2.6.2 below.

⁽¹⁰⁾**N.D.E.** This corollary, used implicitly in VIII, 6.7, has been added; see also VIB, 6.2.5.

⁽¹¹⁾**N.D.E.** For example, they are always complete intersections, cf. VIIB, 5.5.1. Moreover, if $\text{char}(k) = 0$, G is smooth (VIB, 1.6.1; see also VIIB, 3.3.1).

SGA 1, I §9, this amounts to saying that, for every finite extension K of the base field k and every point $\bar{y}$ of $\bar{G} = G \otimes_k K$ above y , $\mathcal{O}_{\bar{G}, \bar{y}}$ is Cohen–Macaulay. If the extension K has been chosen sufficiently large, i.e. if K contains a normal extension of k containing the residue fields $\kappa(y)$ and $\kappa(x)$, then $\bar{y}$ is (strictly) rational over K , as is every point $\bar{x}$ of $\bar{G}$ above x .⁽¹²⁾ Since the automorphism $r_{\bar{x}} \circ (r_{\bar{y}})^{-1}$ sends $\bar{y}$ to $\bar{x}$, it follows that $\mathcal{O}_{\bar{G}, \bar{x}}$, and hence $\mathcal{O}_{G, x}$ (SGA 1, I §9), are Cohen–Macaulay.

When A is again arbitrary, the preceding argument applies to $k \otimes_A G$, so $k \otimes_A \mathcal{O}_{G, x}$ is Cohen–Macaulay. If $a_1, \dots, a_n$ is a sequence of elements of $\mathcal{O}_{G, x}$ whose image in $k \otimes_A \mathcal{O}_{G, x}$ is a system of parameters, it follows from SGA 1, IV 5.7 or EGA 0_{IV}, 15.1.16 that $a_1, \dots, a_n$ is an $\mathcal{O}_{G, x}$ -regular sequence and $\mathcal{O}_{G, x}/(a_1, \dots, a_n)$ is finite flat (hence finite free) over A .

Lemma 1.1.2. *Every nonempty scheme X locally of finite type over an artinian ring A contains a closed point x whose local ring is Cohen–Macaulay.*

One can obviously suppose X affine with algebra B and argue by induction on $\dim X$ (the assertion is clear if X is discrete, all local rings then being artinian). Since B is of finite type over A , if $\dim B > 0$, B contains a noninvertible element a that is not a zero divisor.⁽¹³⁾ The closed subscheme $X' = \operatorname{Spec} B/(a)$ of X then has dimension strictly smaller than $\dim X$, and contains, by the induction hypothesis, a closed point x such that $\mathcal{O}_{X', x}$ is Cohen–Macaulay. Since $\mathcal{O}_{X', x} = \mathcal{O}_{X, x}/(a)$ and a is noninvertible and not a zero divisor in $\mathcal{O}_{X, x}$, $\mathcal{O}_{X, x}$ is Cohen–Macaulay (see also EGA IV₂, 6.11.3).

Proposition 1.2. *Let A be an artinian local ring, G an A -group locally of finite type and flat over A , and x a closed point of G . There then exists a local A -algebra A' , finite and free over A , such that every point of $G \otimes_A A'$ above x is strictly rational over A' .*

⁽¹⁴⁾Indeed, let k_1 be a normal extension of finite degree of k containing the residue field $\kappa(x)$ of x . By Lemma V 4.1.2, there exists a local A -algebra A_1 , finite and free over A , with residue field k_1 . In this case (cf. N.D.E. (11)), all points $g_1, \dots, g_n$ of $G \otimes_A A_1$ above $x \in G$ have residue field k_1 (i.e. $g_1, \dots, g_n$ are rational over A_1 in the sense of Exp. V, §4.e)).

Let, then, $B_1, \dots, B_n$ be the local rings of $g_1, \dots, g_n$. By 1.1.1, $B_1, \dots, B_n$ have quotients $B'_1, \dots, B'_n$ which are artinian and finite free over A_1 . Put $A' = B'_1 \otimes_{A_1} \dots \otimes_{A_1} B'_n$. Then A' is local, finite and free over A_1 , and, for each $i = 1, \dots, n$, one has surjective homomorphisms

$$B_i \otimes_{A_1} A' \twoheadrightarrow B'_i \otimes_{A_1} A' \twoheadrightarrow A',$$

the second being induced by the multiplication map $B'_i \otimes_{A_1} B'_i \rightarrow B'_i$. Consequently A' answers the question.

1.3. — Let e be the neutral element (or origin) of G , that is, the image of the only point of $\operatorname{Spec} A$ under the unit section $\operatorname{Spec} A \rightarrow G$. By the very definition, e is strictly rational over A .

Proposition 1.3.1. ⁽¹⁵⁾ *Let G be a group locally of finite type and flat over an artinian ring A , and let K (resp. $\bar{k}$) be the perfect closure (resp. an algebraic closure) of the residue field k of A .*

⁽¹²⁾**N.D.E.** Indeed, the hypothesis on K implies that, for every extension L of K , every k -morphism $\kappa(x) \rightarrow L$ (resp. $\kappa(y) \rightarrow L$) factors through K ; consequently all points of $G \otimes_k K$ above x or y have residue field K , i.e. are (strictly) rational over K .

⁽¹³⁾**N.D.E.** Indeed, B is a noetherian Jacobson ring (cf. [BAC], V §3.4). If every noninvertible element is a zero divisor, every prime ideal is an associated prime ideal of B , so, in particular, B has only finitely many maximal ideals $\mathfrak{m}_1, \dots, \mathfrak{m}_n$. Since B is a Jacobson ring, the intersection of the $\mathfrak{m}_i$ is the nilradical of B , and it follows that each $\mathfrak{m}_i$ is a minimal prime ideal of B , so $\dim B = 0$.

⁽¹⁴⁾**N.D.E.** The original has been expanded in what follows.

⁽¹⁵⁾**N.D.E.** Point (1), useful in the examples 1.3.2 below, has been added.

(1) For every closed point x of $\overline{G} = G \otimes_k \overline{k}$, the local rings $\mathcal{O}_{\overline{G},e}$ and $\mathcal{O}_{\overline{G},x}$ are isomorphic. In particular, the tangent spaces $T_e \overline{G}$ and $T_x \overline{G}$ have the same dimension.

(2) The following assertions are equivalent:

- (i) $G \otimes_A K$ is reduced.
- (i bis) $\mathcal{O}_{G,e} \otimes_A K$ is reduced.
- (ii) G is smooth over A .
- (ii bis) G is smooth over A at the origin.

PROOF. (1) Let x be a closed point of $\overline{G}$; there is exactly one $\overline{k}$ -morphism $s : \text{Spec } \overline{k} \rightarrow \overline{G}$ whose image is x ; right translation r_s then induces an isomorphism from $\mathcal{O}_{\overline{G},e} = \mathcal{O}_{G,e} \otimes_k \overline{k}$ onto $\mathcal{O}_{\overline{G},x}$, whence assertion (1).

Let us prove assertion (2). By means of SGA 1, II.2.1, one immediately reduces to the case where A is a field ($A = k$). The implications (i) $\Rightarrow$ (i bis), (ii) $\Rightarrow$ (ii bis), (ii) $\Rightarrow$ (i) and (ii bis) $\Rightarrow$ (i bis) are obvious, so it suffices to prove that (i bis) implies (ii).

Now it follows from (i bis) that $\mathcal{O}_{G,e} \otimes_k \overline{k}$ is reduced. Thus, by (1), $\mathcal{O}_{\overline{G},x}$ is reduced for every closed point x of $\overline{G}$, so $\overline{G}$ is reduced. Hence, since $\overline{G}$ is locally of finite type over $\overline{k}$, there exists at least one closed point y such that $\mathcal{O}_{\overline{G},y}$ is regular. Since, by (1), the local rings of the closed points of $\overline{G}$ are all isomorphic to $\mathcal{O}_{\overline{G},e}$, one sees that all these local rings are regular, so $\overline{G}$ is smooth over $\overline{k}$, and therefore G smooth over k . $\square$

⁽¹⁶⁾One can now give the examples below, pointed out by M. Raynaud, of group schemes G over an imperfect field k such that G_{red} is not a k -group scheme.

Examples 1.3.2. Let k be an imperfect field of characteristic $p > 0$, $t \in k - k^p$, $\overline{k}$ an algebraic closure of k , and $\alpha \in \overline{k}$ such that $\alpha^p = t$.

(1) Consider the additive group $\mathbf{G}_{a,k} = \text{Spec } k[X]$, and let G be the subgroup scheme, finite over k , defined by the additive polynomial $X^{p^2} - tX^p$. Then

$$G_{\text{red}} = \text{Spec } k[X]/X(X^{p(p-1)} - t)$$

is étale at the origin. If it were a k -group scheme, it would be smooth (by 1.3.1); now G_{red} is not geometrically reduced, so it is not a k -group scheme.

(2) Consider $\mathbf{G}_{a,k}^4 = \text{Spec } k[X, Y, U, V]$, and let G be the subgroup scheme defined by the ideal I generated by the additive polynomials $P = X^p - tY^p$ and $Q = U^p - tV^p$. Then G has dimension 2 and is irreducible, for $(G_{\overline{k}})_{\text{red}} \simeq \text{Spec } \overline{k}[Y, V]$ is so.

Let $A = k[X, Y, U, V]$ and $\mathfrak{m}$ its augmentation ideal. Denote by x, y, u, v the images of dX, dY, dU, dV in $\Omega_{A/k}^1 \otimes_A (A/\mathfrak{m})$, regarded as linear forms on the tangent space $k^4 = T_0 \mathbf{G}_{a,k}^4$. Let us show that the subspace $E = T_0 G_{\text{red}}$ equals k^4 . Otherwise there would exist a linear form $f = ax + by + a'u + b'v$, with $a, b, a', b' \in k$ not all zero, vanishing on E . Recall that formation of $\Omega_{A/k}^1$ (and therefore of tangent spaces) commutes with base change (cf. EGA IV₄, 16.4.5), and identify f with its image in $(\overline{k}^4)^*$. Since $(G_{\overline{k}})_{\text{red}} \subset (G_{\text{red}})_{\overline{k}}$, f vanishes on the subspace $T_0(G_{\overline{k}})_{\text{red}}$ of $\overline{k}^4$, defined by the equations $g_1 = x - \alpha y$ and $g_2 = u - \alpha v$, and hence $f = \lambda g_1 + \mu g_2$, with $\lambda, \mu \in \overline{k}$. Now $\lambda g_1 + \mu g_2$ belongs to k^4 only if $\lambda = \mu = 0$! This contradiction shows that $E = k^4$, and therefore $T_0(G_{\text{red}})_{\overline{k}} = \overline{k}^4$.

On the other hand, $R = XV - YU$ belongs to $\sqrt{I}$, for $R^p = (X^p - tY^p)V^p - Y^p(U^p - tV^p)$. Consequently, the tangent space F at the point $(\alpha, 1, \alpha, 1)$ of $(G_{\text{red}})_{\overline{k}}$ is contained in the hyperplane H of $\overline{k}^4$ with equation $\alpha dV + dX - dU - \alpha dY = 0$, hence has dimension ≤ 3 .⁽¹⁷⁾ Thus, by point (1) of 1.3.1, G_{red} is not a k -group scheme.

⁽¹⁶⁾**N.D.E.** The following sentence and examples 1.3.2 have been added.

⁽¹⁷⁾**N.D.E.** In fact, one sees without difficulty that $\sqrt{I}$ is generated by P, Q, R at every point $\neq 0$, so $F = H$.

2. Connected components of an A -group locally of finite type

2.1. — First consider any A -group G , and let G' be the connected component of the origin e of G . This connected component is obviously closed, so we can identify it with the reduced closed subscheme of G having G' as its underlying space.⁽¹⁸⁾

Proposition 2.1.1. *For every extension K of the residue field k of A , $G' \otimes_A K$ has as its underlying space the connected component of the origin in the K -group $G \otimes_A K$ (i.e. G' is geometrically connected).*

Indeed, let $(G \otimes_A K)'$ be the connected component of the origin in $G \otimes_A K$. Since the image of $(G \otimes_A K)'$ in G is connected and contains the neutral element of G , this image is contained in G' , so $(G \otimes_A K)'$ is contained in the inverse image $G' \otimes_A K$ of G' in $G \otimes_A K$. The proposition therefore follows from connectedness of $G' \otimes_A K$, proved in Lemma 2.1.2:

Lemma 2.1.2. *Let X and Y be two connected schemes over a field k . If X contains a rational point, $X \times_k Y$ is connected.*

We give below a direct proof of this result of EGA IV₂ (4.5.8 and 4.5.14).

First suppose Y nonempty, connected and affine with algebra B . In this case, $X \times_k Y$ is the spectrum of the quasi-coherent $\mathcal{O}_X$ -algebra $\mathcal{B} = \mathcal{O}_X \otimes_k B$. We want to show that every open, closed and nonempty subset U of $X \times_k Y$ coincides with $X \times_k Y$. Now U is affine over X and has as its affine $\mathcal{O}_X$ -algebra a direct factor of $\mathcal{B}$. It therefore follows from Lemma 2.1.3 below that the image of U in X is open and closed, i.e. coincides with the whole of X . This image contains, in particular, a rational point x of X , so U meets the inverse image of x in $X \times_k Y$. Since the latter is isomorphic to Y , hence connected, U contains this inverse image. The same result would hold for the complement of U in $X \times_k Y$ if U were distinct from $X \times_k Y$, which would be absurd.

If Y is now any k -scheme, what precedes shows that the fibers of the canonical projection $X \times_k Y \rightarrow Y$ are connected. If x is a rational point of X , these fibers all meet the subscheme $\{x\} \times_k Y$, which is itself connected, whence the proposition.

Lemma 2.1.3. *Let X be a scheme and $\mathcal{A}$ a quasi-coherent $\mathcal{O}_X$ -algebra which is locally⁽¹⁹⁾ a direct factor of a free $\mathcal{O}_X$ -module. The image of $\text{Spec } \mathcal{A}$ in X is then open and closed.*

Let V be this image. It is clear that V is contained in the support of $\mathcal{A}$. Conversely, if x belongs to the support of $\mathcal{A}$, $\mathcal{A}_x$ is nonzero and is a free $\mathcal{O}_{X,x}$ -module, since, by Kaplansky, every projective module over a local ring is free. Consequently, the fiber of $\text{Spec } \mathcal{A}$ at x , affine with algebra $\mathcal{A}_x \otimes_{\mathcal{O}_{X,x}} \kappa(x)$, is nonempty. Thus the image of $\text{Spec } \mathcal{A}$ coincides with the support of $\mathcal{A}$.

If s is the unit section of $\mathcal{A}$, the equality $s_x = 0$ implies that s , and hence $\mathcal{A}$, are zero in a neighborhood of x . Thus the support of $\mathcal{A}$ is closed. On the other hand, one can suppose $\mathcal{A}$ a direct factor of the direct sum L of a family $(\mathcal{O}_X^\alpha)$ of copies of $\mathcal{O}_X$.⁽²⁰⁾ For every β , denote by φ^β the restriction to $\mathcal{A}$ of the canonical projection of $L = \bigoplus_\alpha \mathcal{O}_X^\alpha$ onto $\mathcal{O}_X^\beta$. For $x \in X$, denote by $\mathcal{A}_x^*$ the dual module $\text{Hom}_{\mathcal{O}_{X,x}}(\mathcal{A}_x, \mathcal{O}_{X,x})$. Since $\mathcal{A}_x$ is a direct factor of L_x , the restriction map

$$\text{Hom}_{\mathcal{O}_{X,x}}(L_x, \mathcal{O}_{X,x}) \longrightarrow \text{Hom}_{\mathcal{O}_{X,x}}(\mathcal{A}_x, \mathcal{O}_{X,x}) = \mathcal{A}_x^*$$

is surjective. If $\mathcal{A}_x \neq 0$, since $\mathcal{A}_x$ is free and nonzero, there exist $a \in \mathcal{A}_x$ and a linear form $\phi \in \mathcal{A}_x^*$ such that $\phi(a) = 1$. There therefore exists a family (ξ^α) of elements of $\mathcal{O}_{X,x}$ such that, for every $u \in \mathcal{A}_{X,x}$, one has $\phi(u) = \sum_\alpha \xi^\alpha \varphi^\alpha(u)$ (this sum being finite, since $\varphi^\alpha(u) = 0$ except

⁽¹⁸⁾**N.D.E.** One will see below (2.2) that G' is a group in the category $(\text{Sch}/k)_{\text{red}}$.

⁽¹⁹⁾**N.D.E.** The word “locally” has been added. Furthermore, the reference to Kaplansky is: *Projective modules*, Ann. of Maths. **68** (1958), 372–377; see also [BAC], §II.3, Ex. 3.

⁽²⁰⁾**N.D.E.** The original has been expanded in what follows.

for finitely many indices). Applying this to $u = a$, one obtains that there exist $\alpha_1, \dots, \alpha_n$ such that

$$1 = \phi(a) = \xi^{\alpha_1} \varphi^{\alpha_1}(a) + \dots + \xi^{\alpha_n} \varphi^{\alpha_n}(a).$$

There then exists an open neighborhood U of x such that a and the ξ^{α_i} come from sections $\tilde{a} \in \mathcal{A}(U)$ and $\tilde{\xi}^{\alpha_i} \in \mathcal{O}_X(U)$, and the equality $\sum_i \tilde{\xi}^{\alpha_i} \varphi^{\alpha_i}(\tilde{a}) = 1$ on U shows that $\tilde{a}_y \neq 0$ for every $y \in U$. (“The support of a projective module is open.”)

2.2. — The notation still being that of 2.1, it is clear that G' is a reduced k -scheme. Lemma 2.1.2 shows that $G' \times_k G'$ is connected, so $(G' \times_k G')_{\text{red}}$ is the reduced subscheme of $G \times_A G$ whose underlying space is the connected component of the origin. In particular, the multiplication morphism $\mu : G \times_A G \rightarrow G$ induces a morphism $\mu' : (G' \times_k G')_{\text{red}} \rightarrow G'$ making G' a group in $(\text{Sch}/k)_{\text{red}}$.

2.2.bis. — ⁽²¹⁾Recall (cf. Exp. V) that, if P is a scheme, $\underline{P}$ denotes its underlying topological space. One then defines an A -subfunctor G^0 of G by putting, for every A -scheme S ,

$$G^0(S) = \{u \in G(S) \mid \underline{u}(\underline{S}) \subset \underline{G'}\}.$$

Let $c : G \rightarrow G$ be the inversion morphism; since $c(G') = G'$, one has $c \circ u \in G^0(S)$ for every $u \in G^0(S)$. On the other hand, if $u, v \in G^0(S)$, $u \boxtimes v$ sends $\underline{S}$ into the subspace of $G \times_A G$ consisting of the points whose two projections belong to G' ; this subspace is identified with the underlying space of $G' \times_A G'$, connected by Lemma 2.1.2. Consequently, $\mu \circ (u \boxtimes v)$ sends $\underline{S}$ into G' . This shows that G^0 is an A -subgroup functor of G .

If the connected component of e is an open subset of G , the subfunctor G^0 is represented by the subscheme induced by G on this open subset, which is therefore a subgroup scheme of G ; it will also be denoted G^0 . In this case, with the notation of 2.1, one has $G' = (G^0)_{\text{red}}$, and the topological spaces G' and G^0 coincide.⁽²²⁾

2.3. — In accordance with our conventions in 1.1, we again suppose from now on that G is locally of finite type over A . Then G is locally noetherian, hence locally connected,⁽²³⁾ hence: every connected component of G is open.

We shall then denote by G^0 the subscheme induced by G on the connected component G' of the neutral element. By 2.2.bis, G^0 is a subgroup scheme of G which we shall call the neutral component of G ; for every A -scheme S one therefore has:

$$G^0(S) = \{u \in G(S) \mid \underline{u}(\underline{S}) \subset \underline{G^0} = \underline{G'}\}.$$

Let G^α be any connected component of G , and $\nu^\alpha : G^\alpha \times_A G^0 \rightarrow G$ the morphism defined by the equalities

$$\nu^\alpha(S)(g, \gamma) = g\gamma g^{-1}$$

for every $S \in (\text{Sch}/A)$, $g \in G^\alpha(S)$, $\gamma \in G^0(S)$.

If e is the origin of G , the restriction of ν^α to $G^\alpha \times_A \{e\}$ is the zero morphism; since $G^\alpha \times_A G^0$ is connected by 2.1.2, one sees that ν^α factors through G^0 . Thus, for every A -scheme S , $G^0(S)$ is an invariant subgroup of $G(S)$. One has therefore obtained the following proposition:⁽²⁴⁾

⁽²¹⁾**N.D.E.** This paragraph has been added to make the connection with VIB, §3.

⁽²²⁾**N.D.E.** In this exposé, the notation G^0 is reserved for the case where the connected component of e is open; in VIB, §3, this connected component will be denoted G^0 in every case. This is a slight abuse of notation, but is compatible with what precedes when the connected component is the underlying topological space of an open subgroup scheme G^0 of G .

⁽²³⁾**N.D.E.** Indeed, in a noetherian space, the connected components are finite in number, so each is open; see also EGA I, 6.1.9.

⁽²⁴⁾**N.D.E.** The numbering 2.3.1 has been added to bring out this statement. Note further that G^0 is even a characteristic subgroup of G , cf. 2.6.5 (ii).

Proposition 2.3.1. *Let G be an A -group locally of finite type. Then the neutral component G^0 is an open invariant subgroup scheme of G .*

Proposition 2.4. *Let G be an A -group locally of finite type.*

- (i) G^0 is irreducible and $G^0 \otimes_A k$ is geometrically irreducible over k .
- (ii) G^0 is quasi-compact, hence of finite type over A .

(25)

PROOF. (i) Since G^0 and $G^0 \otimes_A k$ have the same underlying topological space, it suffices to show the second assertion. Let $\bar{k}$ be an algebraic closure of k . By 2.2, $(G \otimes_A \bar{k})_{\text{red}}$ is a reduced $\bar{k}$ -group locally of finite type, hence smooth over $\bar{k}$ (1.3.1). A fortiori the local rings of $(G \otimes_A \bar{k})_{\text{red}}$ are integral domains, so, ⁽²⁶⁾ since $G \otimes_A \bar{k}$ is locally noetherian, the connected components of $G \otimes_A \bar{k}$ are irreducible (cf. EGA I, 6.1.10). In particular, the connected component $G^0 \otimes_A \bar{k}$ (cf. 2.1.1) is irreducible.

(ii) Let us now show that G^0 is of finite type over A . Since G^0 is locally of finite type over A , it suffices to prove G^0 quasi-compact. Since G^0 is irreducible, this follows from 0.5.1. $\square$

Corollary 2.4.1. *Every connected component of G is irreducible, ⁽²⁷⁾ of finite type over A , ⁽²⁸⁾ and of the same dimension as G^0 .*

Indeed, one can suppose A equal to its residue field k . Let, then, C be a connected component of G , x a closed point of C , $\kappa(x)$ the residue field of x , and k' a normal extension of k containing $\kappa(x)$ and of finite degree over k . The canonical projection $\pi : C \otimes_k k' \rightarrow C$ is open and closed; ⁽²⁹⁾ consequently, if C' is a connected component of $C \otimes_k k'$, the projection $C' \rightarrow C$ is surjective, so C' contains a point $y \in \pi^{-1}(x)$, and such a point is rational over k' (cf. the proof of 1.2), so C' is the image of $G^0 \otimes_k k'$ under translation r_y . Now $G^0 \otimes_k k'$ is of finite type over k' by 2.4, and $\pi^{-1}(x)$ is finite (of cardinality $\leq [k' : k]$), so $C \otimes_k k'$ is of finite type over k' , and hence C is of finite type over k .

On the other hand, since $G^0 \otimes_k k'$ is irreducible by 2.4, the same holds for C' , ⁽³⁰⁾ and therefore also for C , since the projection $C' \rightarrow C$ is surjective.

Finally, we have seen above that $C \otimes_k k'$ is a disjoint union of finitely many translates of $G^0 \otimes_k k'$. Since dimension is invariant under extension of the base field (cf. EGA IV₂, 4.1.4), it follows that C has the same dimension as G^0 . (Moreover, by EGA IV₂, 5.2.1, one has $\dim_g G = \dim G^0$ for every point $g \in G$.)

2.5. — This paragraph has been added. The results below appear in Exp. VIB, but could (or should) have appeared in VIA, and it is useful to have them already, in order to make Theorem 3.2 below more precise.

Lemma 2.5.1. *Let $(A, \mathfrak{m})$ be an artinian local ring, and $k = A/\mathfrak{m}$ its residue field.*

- (i) *If X is an A -scheme such that $X \otimes_A k$ is locally of finite type (resp. of finite type) over k , then the same holds for X over A .*
- (ii) *Let $u : X \rightarrow Y$ be a morphism of A -schemes. If $u \otimes_A k$ is an immersion (resp. a closed immersion), the same holds for u .*

⁽²⁵⁾**N.D.E.** In the statement, “ G^0 is geometrically irreducible” has been replaced by “ $G^0 \otimes_A k$ is geometrically irreducible over k ”, and the proof has been expanded.

⁽²⁶⁾**N.D.E.** The original has been expanded by adding the reference to EGA I, 6.1.10.

⁽²⁷⁾**N.D.E.** One must beware that a nonneutral connected component is not geometrically connected in general. For example, if $k = \mathbf{R}$, the group $\mu_{3,\mathbf{R}}$, represented by $\mathbf{R}[X]/(X^3 - 1)$, has two connected components: $\{e\} = \text{Spec } \mathbf{R}$ and $C = \text{Spec } \mathbf{R}[X]/(X^2 + X + 1)$, and $C \otimes_{\mathbf{R}} \mathbf{C}$ has two components.

⁽²⁸⁾**N.D.E.** The following assertion has been added, cf. VIB, 1.5.

⁽²⁹⁾**N.D.E.** The original has been expanded in what follows.

⁽³⁰⁾**N.D.E.** The original has been simplified here.

PROOF. (i) Suppose $X \otimes_A k$ locally of finite type over k . Let $U = \operatorname{Spec} B$ be an affine open subset of X . By hypothesis, there exist elements $x_1, \dots, x_n$ of B whose images generate $B/\mathfrak{m}B$ as a k -algebra, and it follows from the “nilpotent Nakayama lemma” that the x_i generate B as an A -algebra. This proves that X is locally of finite type over A . If moreover $X \otimes_A k$ is quasi-compact, the same holds for X (which has the same underlying topological space), and hence X is of finite type over A . This proves (i).

Let us prove (ii). Suppose $u \otimes_A k$ an immersion (resp. a closed immersion). Then u is a homeomorphism from X onto a locally closed (resp. closed) subset of Y and, for every $x \in X$, the ring morphism $\varphi_x : \mathcal{O}_{Y,u(x)} \rightarrow \mathcal{O}_{X,x}$ is such that $\varphi_x \otimes_A k$ is surjective. By the nilpotent Nakayama lemma, it follows that φ_x is surjective, so u is an immersion (resp. a closed immersion). $\square$

Proposition 2.5.2. *Let A be an artinian local ring with residue field k , and let $u : G \rightarrow H$ be a quasi-compact morphism between A -group schemes locally of finite type.*

(a) *The set $u(G)$ is a closed subset of H whose connected components are irreducible and all have the same dimension.*

(b) *One has $\dim G = \dim u(G) + \dim \operatorname{Ker}(u)$.*

(c) *If u is a monomorphism, it is a closed immersion.*

PROOF. By the preceding lemma, it suffices to prove the proposition in the case $A = k$. Moreover, since the properties under consideration are stable under fpqc descent, and dimension is invariant under extension of the base field, one can suppose k algebraically closed.

Let us prove (a). Denote by C the reduced subscheme of H whose underlying topological space is $\overline{u(G)}$. Since $u(G)$ is stable under the inversion morphism of H , so is C . On the other hand, $u : G \rightarrow C$ is quasi-compact and dominant, so, by EGA IV₂, 2.3.7, the same holds for $u \times_k \operatorname{id}_G$ and $\operatorname{id}_H \times_k u$, hence for their composite $u \times_k u : G \times_k G \rightarrow C \times_k C$. Consequently, multiplication in H sends $C \times_k C$ into C , so C is a subgroup scheme of H .

Thus, replacing H by C , one reduces to the case where u is dominant. Then $G(k)$ is dense in H , hence meets every connected component of H , and therefore acts transitively on the set of these connected components. It thus suffices to show that $u(G)$ contains H^0 . Replacing G by $u^{-1}(H^0)$, one can therefore suppose $H = H^0$; in this case, by 2.4, H is irreducible and of finite type over k , hence noetherian. On the other hand, u is locally of finite type (cf. EGA I, 6.6.6) and quasi-compact, hence of finite type. Consequently, by Chevalley’s constructibility theorem (cf. EGA IV₁, 1.8.5), $u(G)$ is a constructible (and dense) subset of $H = \overline{u(G)}$, and therefore contains a dense open subset U of H (cf. EGA 0_{III}, 9.2.2). Then, by 0.5, one has $H = U \cdot U \subset u(G)$, whence $u(G) = H$. Taking account of 2.4.1, this proves assertion (a).

Let us prove (b). First recall that the functor $\operatorname{Ker}(u)$ (cf. I, 2.3.6.1) is represented by $u^{-1}(e)$, where e denotes the neutral element of H . Since u is of finite type, $\operatorname{Ker}(u)$ is of finite type over k . On the other hand, replacing H by the reduced closed subscheme $u(G)$, one can suppose u surjective. Denote by u^0 the restriction of u to G^0 . Since G and $\operatorname{Ker}(u)$ are equidimensional, and $\operatorname{Ker}(u)^0 \subset \operatorname{Ker}(u^0)$, one reduces to the case where G , and hence also H , are irreducible.

Then, by EGA IV₃, 9.2.6.2 and 10.6.1 (ii), the set of $y \in H$ such that $\dim u^{-1}(y) = \dim G - \dim H$ contains a nonempty open subset V . Since u is surjective, $U = u^{-1}(V)$ is then a nonempty open subset of G , and therefore contains a closed point x of G , since G is a Jacobson scheme (cf. EGA IV₃, 10.4.8). Then right translation r_x is an isomorphism from $\operatorname{Ker}(u)$ onto $u^{-1}(u(x))$, whence:

$$\dim \operatorname{Ker}(u) = \dim u^{-1}(u(x)) = \dim G - \dim H.$$

Let us prove (c), following [DG70], I, §3.4. (Another proof is given in Exp. VIB, 1.4.2.) Suppose u a monomorphism. If C is a connected component of G , there exists a closed point $x \in G$ such that $C = r_x(G^0)$, and, denoting by u_C (resp. u^0) the restriction of u to C (resp.

G^0), one has $u_C = r_{u(x)} \circ u^0 \circ r_x^{-1}$; thus it suffices to show that u^0 is a closed immersion. One can therefore suppose $G = G^0$, so G is irreducible and of finite type over k .

Let ξ be the generic point of G ; then $\mathcal{O}_{G,\xi}$ is an artinian local ring; denote its maximal ideal by $\mathfrak{m}$. On the other hand, let $h = u(\xi)$, $\mathfrak{n}$ the maximal ideal of $\mathcal{O}_{H,h}$, and $A = \mathcal{O}_{G,\xi}/\mathfrak{n}\mathcal{O}_{G,\xi}$. Since u is a monomorphism, so is the morphism $u_h : \text{Spec}(A) \rightarrow \text{Spec}(\kappa(h))$ obtained by base change, so the multiplication morphism $A \otimes_{\kappa(h)} A \rightarrow A$ is an isomorphism (cf. EGA I, 5.3.8), whence $A = \kappa(h)$. By Nakayama's lemma (since $\mathfrak{n}\mathcal{O}_{G,\xi}$ is contained in $\mathfrak{m}$, hence nilpotent), it follows that the morphism $\mathcal{O}_{H,h} \rightarrow \mathcal{O}_{G,\xi}$ is surjective.

Let, then, V be an affine open subset of H containing h , U a nonempty affine open subset of G contained in $u^{-1}(V)$, $\varphi : \mathcal{O}_H(V) \rightarrow \mathcal{O}_G(U)$ the morphism of k -algebras induced by u , $\mathfrak{p}$ the prime ideal of $\mathcal{O}_G(U)$ corresponding to η , and $\mathfrak{q} = \varphi^{-1}(\mathfrak{p})$. Since G is of finite type over k , $\mathcal{O}_G(U)$ is generated as a k -algebra by finitely many elements $a_1, \dots, a_n$. By what precedes, there exist elements $b_1, \dots, b_n, s$ in $\mathcal{O}_H(V)$ such that $s \notin \mathfrak{q}$ and the equalities $a_i/1 = \varphi(b_i)/\varphi(s)$ hold in $\mathcal{O}_G(U)_{\mathfrak{p}}$. There therefore exist elements $t_1, \dots, t_n$ of $\mathcal{O}_G(U) - \mathfrak{p}$ such that $t_i(a_i\varphi(s) - \varphi(b_i)) = 0$. Then, putting $t = t_1 \cdots t_n\varphi(s) \in \mathcal{O}_G(U) - \mathfrak{p}$, the equalities $a_i/1 = \varphi(b_i)/\varphi(s)$ already hold in $\mathcal{O}_G(U)_t$, and, since $t \in \mathcal{O}_G(U) = k[a_1, \dots, a_n]$, there exists $b \in \mathcal{O}_H(V)$ such that $t/1 = \varphi(b)/\varphi(s)^r$ for some $r \in \mathbb{N}$. Consequently, φ induces a surjection from $\mathcal{O}_H(V)_{sb}$ onto $\mathcal{O}_G(U)_t$, and hence u is a local immersion at ξ .

The open subset W of G consisting of the points where u is a local immersion is therefore nonempty. Since G is a Jacobson scheme, W contains a closed point y , and, to show that $W = G$, it suffices to show that every closed point of G belongs to W . Now every closed point x is the image of y under translation $r_x \circ r_y^{-1}$, hence belongs to W , whence $W = G$. This proves that u is a local immersion.

Since G is irreducible, it follows that u is an immersion. Indeed, for every $x \in G$, let U_x and V_x be open subsets of G and H such that $x \in U_x$ and u induces a closed immersion from U_x into V_x . Since U_x is dense in G , $u(U_x)$ is dense in $u(G) \cap V_x$, and since $u(U_x)$ is closed in V_x , one therefore has $u(U_x) = V_x$. Since moreover u is injective, one has $U_x = u^{-1}(V_x)$, and it follows that u induces a closed immersion from G into the open subscheme of H covered by the V_x . Thus $u : G \rightarrow H$ is an immersion. But we have already seen that $u(G)$ is a closed subset of H , so u is a closed immersion. $\square$

Lemma 2.5.3. ⁽³¹⁾ *Let A be an artinian local ring, k its residue field, G a flat A -group, X an A -scheme with a left action $\mu : G \times_A X \rightarrow X$ of G and a section $s_0 : \text{Spec } A \rightarrow X$. (This is the case, for example, if G' is a second A -group and one is given a morphism of A -groups $G \rightarrow G'$.)*

Let φ be the morphism $\mu \circ (\text{id}_G \times s_0)$ from $G = G \times_A A$ to X . If φ is flat at one point g of G , then φ is flat.

PROOF. Since G is flat over A , by the fiberwise flatness criterion (EGA IV₃, 11.3.10.2), it suffices to show that $\varphi \otimes_A k$ is flat, so one can suppose $A = k$. In this case, giving s_0 amounts to giving a k -point $x_0 \in X(k)$, and φ is the morphism $h \mapsto hx_0$.

Let, then, $h \in G$; let us show that φ is flat at h . Let K be an extension of k containing a copy of $\kappa(g)$ and $\kappa(h)$; one has a cartesian square

$$\begin{array}{ccc} G_K & \xrightarrow{\varphi_K} & X_K \\ \downarrow & & \downarrow \\ G & \xrightarrow{\varphi} & X \end{array}$$

in which the two vertical arrows are faithfully flat. Thus, by V 7.4 (i), φ_K is flat at every point $g' \in G_K$ above g , and, to show that φ is flat at h , it suffices to show that φ_K is flat at

⁽³¹⁾**N.D.E.** No finiteness hypotheses are made in this statement; this will be useful below (cf. 6.2 and VIB, §12).

a point h' above h . One is therefore reduced to the case where g and h are rational. Let, then, $u = hg^{-1}$, and let ℓ_u (resp. μ_u) be the left translation of G (resp. X) defined by u ; since $\varphi \circ \ell_u = \mu_u \circ \varphi$, one obtains a commutative square

$$\begin{array}{ccc} \mathcal{O}_{G,h} & \xrightarrow{\sim} & \mathcal{O}_{G,g} \\ \uparrow & & \uparrow \\ \mathcal{O}_{X,hx_0} & \xrightarrow{\sim} & \mathcal{O}_{X,gx_0} \end{array}$$

in which the horizontal arrows are isomorphisms. Since the morphism $\mathcal{O}_{X,gx_0} \rightarrow \mathcal{O}_{G,g}$ is flat, the morphism $\mathcal{O}_{X,hx_0} \rightarrow \mathcal{O}_{G,h}$ is also so. $\square$

Proposition 2.5.4. *Let A be an artinian local ring, k its residue field, G an A -group locally of finite type, and $X \neq \emptyset$ an A -scheme locally of finite type with a left action of G . Suppose the morphism $\varphi : G \times_A X \rightarrow X \times_A X$, defined set-theoretically by $(g, x) \mapsto (gx, x)$, is surjective. Then:*

- (i) *The connected components of X are of finite type, irreducible and all of the same dimension.*
- (ii) *More precisely, let $\bar{k}$ be an algebraic closure of k and x a closed point of $X \otimes_A \bar{k}$; its stabilizer F is a closed subgroup scheme of $G \otimes_A \bar{k}$, and the dimension of the irreducible components of X is $\dim G - \dim F$.*

Taking account of Lemma 2.5.1, one can suppose $A = k$. First suppose k algebraically closed. Then G_{red} is a k -group locally of finite type, and therefore, replacing G by G_{red} and X by X_{red} , one can suppose G and X reduced.

Since $G \times_k X$ is locally of finite type over k , φ is locally of finite type (cf. EGA I, 6.6.6 (v)), hence locally of finite presentation since $X \times_k X$ is locally noetherian. Let x be a rational point of X ; then the morphism $\varphi_x : G \rightarrow X$ obtained from φ by base change is surjective and locally of finite presentation. If η is a maximal point of X , $\mathcal{O}_{X,\eta}$ is a field (since X is reduced), hence φ_x is flat at every point of G above η . Thus, by Lemma 2.5.3, φ_x is flat. Consequently, $\varphi_x : G \rightarrow X$ is faithfully flat and locally of finite presentation, hence open (cf. EGA IV₂, 2.4.6). Since G^0 is open in G , irreducible and quasi-compact (by 2.4), each orbit $G^0x = \varphi_x(G^0)$, as x ranges over the rational points of X , is an irreducible quasi-compact open subset of X , hence of finite type over k (since X is locally of finite type over k).

Since every nonempty open subset of X contains a rational point, it follows that X is covered by these open subsets. Moreover, two such open subsets are either disjoint or equal. Indeed, if $\varphi_x(G^0) \cap \varphi_y(G^0)$ is nonempty, it contains a rational point z , so there exist two rational points $g, h \in G^0$ such that $gx = z = hy$, whence $x = g^{-1}z$ and $y = h^{-1}z$, and hence $\varphi_x(G^0) = \varphi_z(G^0) = \varphi_y(G^0)$. It follows that the orbits $\varphi_x(G^0)$ are also closed, and are therefore both the connected components and the irreducible components of X .

Finally, let x, y be two rational points of X . Since φ_x is surjective, there exists a rational point $g \in G$ such that $y = gx$, and, since G^0 is an invariant subgroup of G , the orbit G^0y is the image of G^0 under translation ℓ_g of X , so G^0y and G^0x have the same dimension.

Moreover, by I, 2.3.3.1, the stabilizer of x is represented by the closed subscheme F of G defined by the cartesian square below:

$$\begin{array}{ccc} F & \longrightarrow & G \\ \downarrow & & \downarrow \varphi_x \\ \text{Spec } k & \longrightarrow & X. \end{array}$$

Then F is a k -group locally of finite type, $F \cap G^0$ is a k -group of finite type containing F^0 , and, by 2.4.1, F and $F \cap G^0$ are equidimensional, of the same dimension as F^0 . Let $C = \varphi_x(G^0)$ be the irreducible component of X containing x . Proceeding as in the proof of point (b) of 2.5.2, one obtains $\dim C = \dim G^0 - \dim F^0 = \dim G - \dim F$.

In the general case (i.e. for any field k), let $\bar{k}$ be an algebraic closure of k . Let C be a connected component of X and C' a connected component of $C \otimes_k \bar{k}$; then C' is a connected component of $X' = X \otimes_k \bar{k}$. The morphism $\pi : X' \rightarrow X$ is open (cf. EGA IV₂, 2.4.10), and, being integral, it is also closed; consequently $\pi(C') = C$. Since C' is irreducible and quasi-compact, C is irreducible and quasi-compact, hence of finite type over k (since X is locally of finite type over k).

Finally, since dimension is invariant under extension of the base field (cf. EGA IV₂, 4.1.4), $\dim C = \dim C'$, and, since all irreducible components of X' have the same dimension, the same holds for those of X .

2.6. — Supplements. — This paragraph, taken from [Per75] II, §§1–2, with supplements due to O. Gabber, has been added.⁽³²⁾ It shows that the preceding results hold for every group scheme G over a field k . (This will be used in sections 5, 6 and 7 of Exp. VIB.)

Fix a field k . Let us begin with the following lemma (loc. cit., II 2.1.1), which does not appear explicitly in EGA IV₂, §4.4 (although one can doubtless read it between the lines at the beginning of loc. cit., §4.4.1).

Lemma 2.6.0. *Let X be an irreducible k -scheme, K an extension of k , and X' an irreducible component of X_K . The projection $X' \rightarrow X$ is surjective.*

Indeed, let B be a transcendence basis of K over k and $L = k(B) \subset K$. By EGA IV₂, 4.3.2 and 4.4.1, X_L is irreducible and X' dominates X_L . The morphism $X' \rightarrow X_L$ is therefore integral and dominant, hence surjective. Since $X_L \rightarrow X$ is surjective (loc. cit., 4.4.1), $X' \rightarrow X$ is also so.

For the remainder of 2.6, fix a k -group scheme G and an action $\mu : G \times X \rightarrow X$ of G on a k -scheme X satisfying the following condition:

($\star$) the morphism $\Phi : G \times X \rightarrow X \times X$, $(g, x) \mapsto (gx, x)$ is surjective

(this is the case, in particular, for G acting on itself by left translations). For brevity, one will then say: “Let X be a G -scheme satisfying ($\star$)”. Finally, denote by k' the perfect closure of k .

Proposition 2.6.1. *Let X be a G -scheme satisfying ($\star$).*

- (i) *X is geometrically pointwise irreducible over k , i.e. for every extension K of k , every $x \in X_K$ belongs to a unique irreducible component of X_K .*
- (ii) *Every local ring of $(X_{k'})_{\text{red}}$ is normal.*
- (iii) *Let η be a maximal point of $(X_{k'})_{\text{red}}$, $C = \overline{\{\eta\}}$, and L the algebraic closure of k in $\kappa(\eta)$. Then C is an L -scheme and is geometrically irreducible over L (i.e. $C \otimes_L K$ is irreducible for every extension K of L).*
- (iv) *In particular, if x is a rational point of X , the irreducible component of X containing x is geometrically irreducible over k .*

PROOF. (i) Since hypothesis ($\star$) is preserved by every base change $k \rightarrow K$, it suffices to show that every $x \in X$ belongs to a unique irreducible component of X . Since the morphism $\text{Spec}(k') \rightarrow \text{Spec}(k)$ is a universal homeomorphism, one can further suppose k perfect. One can then suppose G and X reduced. Let η be a maximal point of X and z any point of $Z = \overline{\{\eta\}}$. Since X is reduced, the local ring $\mathcal{O}_{X,\eta}$ equals $\kappa(\eta)$, and, since k is perfect, for every extension K of k , $\kappa(\eta) \otimes_k K$ is normal (cf. EGA IV₂, 6.14.2), so every point of X_K above η is normal.

Since Φ is surjective, there exists a point γ of $G \times X$ such that $\Phi(\gamma)$ has projections z and η . Let $K = \kappa(\gamma)$; there then exist rational points g and η' of G_K and X_K such that η' lies above η and $z' = g\eta'$ above z . By what precedes, η' is a normal point of X_K , hence so is

⁽³²⁾**N.D.E.** These results were communicated to us by O. Gabber, in particular 2.6.6, which plays an important role in section 5 of VIB.

z' . Since $\pi : X_K \rightarrow X$ is flat, it follows that $z = \pi(z')$ is a normal point of X (cf. EGA IV₂, 2.1.13). This proves (ii) and (i).

The first assertion of (iii) then follows from (ii). Next, since L is algebraically closed in $\kappa(\eta)$, C is geometrically irreducible over L , by EGA IV₂, 4.5.9.

Finally, if X has a rational point x , it follows from (iii) that $L = k$, and hence X is geometrically irreducible over k . This can also be seen directly as follows (cf. [Per75], II 2.1): let C be the irreducible component of X containing x , and let K be an extension of k ; X_K has a unique point e_K above e , and, by (i), e_K belongs to a unique irreducible component C' of X_K ; on the other hand, by 2.6.0, every irreducible component of C_K contains e_K , hence equals C' . $\square$

Notation. — Temporarily denote by C^0 the reduced closed subscheme of G whose underlying space is the unique irreducible component of G containing the neutral element e .

Corollary 2.6.2. *C^0 is geometrically irreducible over k and is stable set-theoretically under the group law, i.e. $(C_{k'}^0)_{\text{red}}$ is a subgroup of $(G_{k'})_{\text{red}}$. Consequently C^0 is quasi-compact.*

Indeed, by 2.6.1, C^0 is geometrically irreducible over k , so $C^0 \times C^0$ is irreducible, hence $\nu(C^0 \times C^0) \subset C^0$, where ν denotes the morphism $(g, h) \mapsto gh^{-1}$. Since $\text{Spec } k' \rightarrow \text{Spec } k$ is a universal homeomorphism, one has the same conclusion for $C_{k'}^0$, then for $H = (C_{k'}^0)_{\text{red}}$, and hence, since $H \times_{k'} H$ is reduced, ν induces a morphism $H \times_{k'} H \rightarrow H$, i.e. H is a subgroup of $(G_{k'})_{\text{red}}$. Consequently, by 0.5.1, H (and therefore also C^0) is quasi-compact.

Recall 2.6.3. Let Y be a scheme. Recall (EGA 0_{III}, 9.1.1) that a subset E of Y is said to be retrocompact if the inclusion $E \hookrightarrow Y$ is quasi-compact, and that, by EGA IV₁, 1.9.5 (v) and 1.10.1, if U is a retrocompact open subset of Y , the closure $\overline{U}$ of U is the union of the closures $\overline{\{y\}}$ of the points $y \in U$, and of course it suffices to let y range over the maximal points of U , i.e. the maximal points of Y contained in U .

Consequently, if Y is quasi-separated and η a maximal point of Y , the intersection of the closed neighborhoods of η equals $\overline{\{\eta\}}$: indeed, if $y \in Y - \overline{\{\eta\}}$, y is contained in an affine open subset V not containing η ; since Y is quasi-separated, V is retrocompact, so $\overline{V}$ is the union of the $\overline{\{\xi\}}$ as ξ ranges over the maximal points of Y belonging to V , and therefore $\eta \notin \overline{V}$, i.e. $Y - \overline{V}$ is a closed neighborhood of η not containing y .

Proposition 2.6.4. *Let X be a G -scheme satisfying $(\star)$, and let U be an open subset of X .*

- (i) *C^0U is an open subset of X , equal to the union of the irreducible components of X whose generic point belongs to U .*
- (i') *Consequently, if X is irreducible, it is quasi-compact.*
- (ii) *If moreover U is retrocompact in X , C^0U equals $\overline{U}$, hence is an open and closed subset of X .*

PROOF. (i) First, C^0U is open, since it is the union, for $g \in C^0$, of the projections of the open subsets $g \cdot U_{\kappa(g)} \subset X_{\kappa(g)}$, and each projection $X_{\kappa(g)} \rightarrow X$ is open.

To prove the second assertion of (i), one can replace k by k' (since $\text{Spec } k' \rightarrow \text{Spec } k$ is a universal homeomorphism), hence suppose k perfect. One can then suppose G and X reduced, hence geometrically reduced.

Let η be a maximal point of X contained in U , and Z its closure. Consider the morphism $\mu' : C^0 \times Z \rightarrow X$. Since C^0 is geometrically irreducible, $C^0 \times Z$ is irreducible; denote its generic point by γ . Since μ' sends the point $\varepsilon(\eta)$ (where ε denotes the unit section of C^0) to η , γ is sent to a generalization of η , hence to η . Thus μ' sends the underlying space of $C^0 \times Z$ into Z , and hence, since $C^0 \times Z$ is reduced, μ' factors through Z .

Now let $z \in Z$. Put $K = \kappa(z)$; then the morphism $\mu_z : G_K \rightarrow X_K$, $h \mapsto h \cdot z$, is surjective; let α be a maximal point of X_K ; the local ring $\mathcal{O}_{X_K, \alpha}$ is a field, since X_K is reduced, so μ_z is flat at every point of G_K above α , and therefore μ_z is flat by Lemma 2.5.3.

On the other hand, μ_z sends the generic point ω of C_K^0 to a point $t \in Z_K$. Let β be a maximal point of Z_K such that $t \in \overline{\{\beta\}}$; since μ_z is flat, there exists a generalization ξ of ω such that $\mu_z(\xi) = \beta$, and, since ω is a maximal point of G_K , necessarily $\xi = \omega$, so $\mu_z(\omega)$ equals β , which lies above η (since $Z_K \rightarrow Z$ is flat).

Put $L = \kappa(\omega)$ and let ω_L, z_L be the L -points obtained from ω, z ; then $\omega_L \cdot z_L = \beta'$ is a point of Z_L above β , and therefore $z_L = \omega_L^{-1} \cdot \beta' \in C_L^0 \cdot U_L$, whence $z \in C^0 \cdot U$. This proves (i).

Since C^0 is quasi-compact by 2.6.2, point (i') follows: if X is irreducible and U a nonempty affine open subset, X equals $C^0 U$, i.e. is the image of the morphism $C^0 \times U \rightarrow X$, hence is quasi-compact. Finally, if U is retrocompact in X , by 2.6.3, $\overline{U}$ is the union of the $\{\eta\}$ as η ranges over the maximal points of X contained in U , hence equals $C^0 U$. This proves (ii). $\square$

One then obtains the following result ([Per75] II Th. 2.4; see also [Per76], Prop. 4.1.1):

Theorem 2.6.5. *Let k be a field and G a k -group scheme.*

- (i) *There exists a unique subgroup scheme G^0 of G , called the neutral component of G , such that:*
 - (a) *The underlying space of G^0 is the irreducible component of the neutral element.*
 - (b) *$G^0 \rightarrow G$ is a flat closed immersion, i.e. $\mathcal{O}_{G,g} = \mathcal{O}_{G^0,g}$ for every $g \in G^0$.*
- (ii) *Moreover, G^0 is quasi-compact, geometrically irreducible, and a characteristic subgroup of G .*
- (iii) *If G is connected, then $G = G^0$.*

PROOF. (i) First recall that G is separated (0.3), hence a fortiori quasi-separated. Let U be an affine open subset of G containing the generic point ω of C^0 . By 2.6.3 and 2.6.4, $\overline{U} = C^0 U$ is both open and closed, and $C^0 = \overline{\{\omega\}}$ is set-theoretically the intersection of these open and closed subsets.

As U ranges over the affine open subsets containing ω , one obtains a projective system of G -schemes $\overline{U}$, whose transition morphisms are affine (since they are closed immersions). One can therefore form its projective limit G^0 (cf. EGA IV₃, 8.2.2), i.e. for every affine open subset V of G , $G^0 \cap V$ is the spectrum of the algebra

$$\varinjlim_{\overline{U}} \mathcal{O}_G(V \cap \overline{U}) = \mathcal{O}_G(V) / \sum_{\overline{U}} I_{\overline{U}}(V),$$

where $I_{\overline{U}}(V)$ denotes the kernel of $\mathcal{O}_G(V) \rightarrow \mathcal{O}_G(V \cap \overline{U})$. It follows that G^0 has underlying space C^0 and $G^0 \rightarrow G$ is a closed immersion. Moreover, for every $g \in G^0$, $\mathcal{O}_{G^0,g}$ is the inductive limit, as V ranges over the affine open subsets of G containing g , of the k -algebras $\mathcal{O}_{G^0}(V \cap G^0) = \varinjlim_{\overline{U}} \mathcal{O}_G(V \cap \overline{U})$, and this double inductive limit is identified with

$$\varinjlim_{\overline{U}} \varinjlim_{\substack{V \\ g \in V \subset \overline{U}}} \mathcal{O}_G(V) = \mathcal{O}_{G,g},$$

i.e. one has $\mathcal{O}_{G^0,g} = \mathcal{O}_{G,g}$ (see also EGA IV₂, 5.13.3 (ii)). Thus $i : G^0 \rightarrow G$ is a flat closed immersion. Conversely, this condition implies $i^*(\mathcal{O}_G) = \mathcal{O}_{G^0}$, and hence G^0 is uniquely determined by conditions (a) and (b). This proves (i).

The first two assertions of (ii) follow from 2.6.2. Finally, let S be a k -scheme and φ an automorphism of the S -group G_S . For every $s \in S$, φ_s sends $G_{\kappa(s)}^0$ into itself, hence $\varphi(G_S^0) \subset G_S^0$. Moreover, the closed immersion $i_S : G_S^0 \hookrightarrow G_S$ obtained from i by base change is flat, so $\mathcal{O}_{G_S, \varphi(z)} = \mathcal{O}_{G_S^0, \varphi(z)}$ for every $z \in G_S^0$, and hence $\varphi \circ i_S$ factors through G_S^0 . This proves that G^0 is a characteristic subgroup of G , whence (ii). Finally, (iii) is a special case of point (i) of the following proposition. $\square$

(33)

Proposition 2.6.6. *Let k be a field and G a k -group acting on a k -scheme X so that the morphism $G \times X \rightarrow X \times X$, $(g, x) \mapsto (gx, x)$, is surjective. Suppose X quasi-separated. Then:*

- (i) *Every connected component C of X is irreducible.*
- (ii) *Let η be the generic point of C and L the algebraic closure of k in $\kappa(\eta)$. Then C is an L -scheme, geometrically irreducible over L , and the morphism*

$$G_L^0 \times_L C \rightarrow C \times_L C$$

is surjective.

- (iii) *In particular, if C contains a rational point x , C is geometrically irreducible over k , and the morphism $\varphi : G^0 \rightarrow C$, $g \mapsto gx$, is surjective.*

PROOF. (i) Let C be a connected component of X , Y an irreducible component of X contained in C , η the generic point of Y , and U an affine open subset of X containing η . Since X is quasi-separated, U is retrocompact in X , so, by 2.6.4, $\bar{U} = G^0 U$ is an open and closed subset of X meeting C , hence contains C . Now, by 2.6.3, the intersection of the $\bar{U}$, as U ranges over a fundamental system of affine open neighborhoods of η , equals $\{\eta\}$. It follows that $C = \{\eta\}$. This proves (i).

The first assertion of (ii) (and also of (iii)) follows from 2.6.1. Let us begin by showing the second assertion of (iii). Denote by ω (resp. η) the generic point of G^0 (resp. C). Let $z \in C$ and $K = \kappa(z)$. Since G^0 (resp. C) is geometrically irreducible over k , G_K^0 (resp. C_K) is irreducible; denote its generic point by ξ (resp. β). We have seen in the proof of 2.6.4 that the morphism $\mu_z : G_K \rightarrow C_K$, $h \mapsto h \cdot z$, sends ξ to β , and similarly $\varphi(\omega) = \eta$.

Let $L = \kappa(\xi)$ and let ξ_L, z_L be the L -points obtained from ξ, z ; then $\xi_L \cdot z_L = \beta'$ is a point of C_L above $\beta \in C_K$, hence also above $\eta \in C$. Consider the cartesian square:

$$\begin{array}{ccc} G_L^0 & \xrightarrow{\varphi_L} & C_L \\ \pi_{G^0} \downarrow & & \downarrow \pi_C \\ G^0 & \xrightarrow{\varphi} & C, \end{array}$$

since $\varphi_L(\pi_{G^0}^{-1}(\omega)) = \pi_C^{-1}(\eta)$ (cf. EGA I, 3.4.8), there exists $g \in G_L^0$ such that $\varphi_L(g) = \beta'$. Thus $z_L = \xi_L^{-1} \cdot \varphi_L(g) = \varphi_L(\xi_L^{-1}g)$, whence $\varphi(\pi_{G^0}(\xi_L^{-1}g)) = \pi_C(z_L) = z$. This proves that φ is surjective.

Now let us prove the second assertion of (ii). It suffices to show that, for every $z \in C$, the morphism $\mu_z : G_L^0 \otimes_L \kappa(z) \rightarrow C \otimes_L \kappa(z)$ is surjective, but this follows from (iii), since z is a rational point of $C \otimes_L \kappa(z)$. $\square$

Remark 2.6.7. Under the hypotheses of 2.6.6, if $C(k) = \emptyset$, the morphism $G^0 \times_k C \rightarrow C \times_k C$ is not necessarily surjective. For example, for $k = \mathbf{R}$ and $G = \{\pm 1\}_{\mathbf{R}}$, the G -torsor $X = \text{Spec } \mathbf{R}[X]/(X^2 + 1)$ is connected, but the morphism $G^0 \times_{\mathbf{R}} X \rightarrow X \times_{\mathbf{R}} X$ is not surjective. (But one has $L = \mathbf{C}$, and the morphism $G^0 \times_{\mathbf{R}} X \rightarrow X \times_{\mathbf{C}} X$ is an isomorphism.)

3. Construction of quotients $F \backslash G$ (for G, F of finite type)

3.1. — Let A be an artinian local ring and $u : F \rightarrow G$ a homomorphism of A -groups. If $\mu : F \times_A F \rightarrow F$ and $\nu : G \times_A G \rightarrow G$ denote the multiplication morphisms, and λ the composite morphism

$$F \times_A G \xrightarrow{u \times G} G \times_A G \xrightarrow{\nu} G,$$

(33) **N.D.E.** This proposition (as well as the preceding results) was communicated to us by O. Gabber; it will be used to correct the proof of Theorem 5.3 of VIB.

recall that the left quotient $F \backslash G$ of G by F is the cokernel of the (Sch/A) -groupoid G_* described below:

$$F \times_A F \times_A G \begin{array}{c} \xrightarrow{F \times \lambda} \\ \xrightarrow[\text{pr}_{2,3}]{\mu \times G} \end{array} F \times_A G \xrightarrow[\text{pr}_2]{\lambda} G$$

(pr_2 and $\text{pr}_{2,3}$ are the projections of $F \times_A G$ and $F \times_A (F \times_A G)$ onto the second factors). We shall say that G_* is the groupoid with base G defined by u (cf. Exp. V, §2.a; as in Exposé V, we do not follow in this exposé the convention of IV, 4.6.15).

Since the unique A -morphism $F \rightarrow \text{Spec } A$ is universally open (EGA IV₂, 2.4.9), pr_2 is an open morphism; the same therefore holds for λ , the composite of pr_2 and the automorphism σ of $F \times_A G$ defined by the following formulas: $\sigma(S)(x, y) = (x, u(S)(x) \cdot y)$, where S is a varying A -scheme and x, y belong to $F(S), G(S)$. One sees in the same way that pr_2 and λ are flat when F is flat over A .

To finish these preliminaries, note also that every A -morphism $s : \text{Spec } A \rightarrow G$ defines an automorphism of the groupoid G_* inducing on G , $F \times_A G$ and $F \times_A F \times_A G$ the automorphisms r_s , $\text{id}_F \times_A r_s$ and $\text{id}_F \times_A \text{id}_F \times_A r_s$ respectively. We shall again denote this automorphism of G_* by r_s , and shall say that r_s is the right translation defined by s (cf. 0.4).

Theorem 3.2. *Let F and G be flat groups locally of finite type over an artinian local ring A . Let $u : F \rightarrow G$ be a quasi-compact homomorphism of A -groups with kernel finite over A . Then.⁽³⁴⁾*

- (i) *The left quotient $F \backslash G$ of G by F exists in (Sch/A) , and the sequence*

$$F \times_A G \xrightarrow[\text{pr}_2]{\lambda} G \xrightarrow{p} F \backslash G$$

is exact in the category of all ringed spaces.

- (ii) *The canonical morphism $p : G \rightarrow F \backslash G$ is surjective and open, and $F \backslash G$ is a direct sum of schemes of finite type over A .*
(ii') *More precisely, $X = F \backslash G$ has a right action of G such that $p(e) \cdot g = p(g)$ for every $g \in G$; consequently, the connected components of X are of finite type over A , irreducible and all of dimension $\dim G - \dim F$.*
(iii) *The canonical morphism $F \times_A G \rightarrow G \times_{(F \backslash G)} G$ is surjective.*
(iv) *If u is a monomorphism,⁽³⁵⁾ then:*
(a) *$F \times_A G \xrightarrow{(\lambda, \text{pr}_2)} G \times_{(F \backslash G)} G$ is an isomorphism, and $G \rightarrow F \backslash G$ is faithfully flat and locally of finite presentation.*
(a') *$F \backslash G$ represents the quotient fppf sheaf $\widetilde{F \backslash G}$, and $G \rightarrow F \backslash G$ is an F -torsor locally trivial for the fppf topology.*
(b) *$F \backslash G$ is flat over A , and is smooth over A if G is.*
(c) *$u : F \rightarrow G$ is a closed immersion, and $F \backslash G$ is separated.*
(d) *If moreover F is an invariant subgroup of G , there exists one and only one A -group structure on $F \backslash G$ such that $p : G \rightarrow F \backslash G$ is a morphism of A -groups.*

In the proof of this theorem, A' will denote a local A -algebra finite and free over A . If R is a relation involving A' , we shall say that “ $R(A')$ is true when A' is sufficiently large” if there exists an algebra A_1 , local, finite and free over A , such that the relation $R(A')$ holds for every algebra A' , local, finite and free over A_1 .

We shall first prove the theorem when F and G are of finite type over A .

3.2.1. — Suppose for a moment that every point of G has a saturated open neighborhood W such that the groupoid induced by G_* on W has a quasi-section (cf. V §6). Then, by V 6.1,

⁽³⁴⁾**N.D.E.** Point (ii') has been added and point (iv) expanded, taking account of the additions made in 2.5.2, 2.5.4 and Exp. V, 6.1.

⁽³⁵⁾**N.D.E.** In this case, the hypothesis that G is flat can be dropped, cf. subsection 3.3.

one has assertions (i), (ii), (iii) and (iv)(a), and $F \backslash G$ is of finite type over k . Moreover, under the hypothesis of (iv), since $G \rightarrow F \backslash G$ is faithfully flat and locally of finite presentation, assertion (b) follows from EGA IV, 2.2.14 and 17.7.7. On the other hand, assertion (iv)(a') follows from (iv)(a), by Exp. IV, 3.4.3.1, 5.2.2 and 5.1.6. Finally, we shall prove (iv)(c) in 3.2.5, and (ii') and (iv)(d) in section 5.

Let us now show the following assertion:

(†) every finite subset of $F \backslash G$ is then contained in an affine open subset.

(36) If U is a quasi-section of the groupoid induced by G_* on a saturated open subset W of G , $U \otimes_A A'$ is a quasi-section of the (Sch/A') -groupoid induced by $G_* \otimes_A A'$ on $W \otimes_A A'$. Moreover, if U_* is the (Sch/A) -groupoid induced by G_* on U , $U_* \otimes_A A'$ is identified with the (Sch/A') -groupoid induced by $G_* \otimes_A A'$ on $U \otimes_A A'$.⁽³⁷⁾

It then follows from the proofs in Exposé V that construction of the quotient $X = F \backslash G$ commutes with an extension $A \rightarrow A'$ of the base of the type considered here.⁽³⁸⁾

Let, then, $x_1, \dots, x_n$ be points of $X = F \backslash G$, which we can suppose closed,⁽³⁹⁾ and $g_1, \dots, g_n$ closed points of G projecting to $x_1, \dots, x_n$. Let V be an everywhere dense affine open subset of X ,⁽⁴⁰⁾ and U its inverse image in G . By 1.2, there exists a local A -algebra A' , finite and free over A , such that the points $g'_1, \dots, g'_p$ of $G' = G \otimes_A A'$ above $g_1, \dots, g_n$ are strictly rational over A' .⁽⁴¹⁾ Since the morphisms $G' \rightarrow G$ and $G \rightarrow X$ are open, $U' = U \otimes_A A'$ is dense in G' , so the open subset $\bigcap_{i=1}^p (U')^{-1} \cdot g'_i$ is nonempty, and therefore contains a closed point x . Thus, by 1.2 (and 0.4.1), one can suppose, enlarging A' if necessary, that x is strictly rational over A' . Then, since $x \in (U')^{-1} \cdot g'_i$, one has $g'_i \in U' \cdot x$.

Denote by V' the inverse image of V in $X' = X \otimes_A A'$; it is an affine open subset of X' , and also the image of U' under the projection $G' \rightarrow X'$. Since right translation r_x is an automorphism of the groupoid $G_* \otimes_A A'$, it induces an automorphism, again denoted r_x , of the quotient X' . Consequently, the image $V' \cdot x = r_x(V')$ of $U' \cdot x$ in X' is an affine open subset of X' containing the images $x'_1, \dots, x'_p$ of $g'_1, \dots, g'_p$.

Now consider the equivalence relation on $X' = X \otimes_A A'$ defined by the projection $X \otimes_A A' \rightarrow X$:

$$X \otimes_A A' \otimes_A A' \begin{matrix} \xrightarrow{d_1} \\ \xrightarrow{d_0} \end{matrix} X \otimes_A A' \longrightarrow X,$$

where d_0 and d_1 are induced by the two canonical injections of A' into $A' \otimes_A A'$. Since A' is a finite free A -algebra, say of rank n , d_0 and d_1 are finite locally free of rank n ; consequently, one can apply the argument in Exp. V, 5.b (taken from the proof of SGA 1, VIII.7.6). One thus obtains that $x'_1, \dots, x'_p$ are contained in a saturated affine open subset W' contained in the affine open subset $V' \cdot x$. The image of W' in X then contains $x_1, \dots, x_n$ and is an affine open subset of X , by V, 4.1 (ii).

(36) **N.D.E.** Let us mention here that if $A = k$ is a field, every quasi-compact open subset of $F \backslash G$ is quasi-projective (a result due to Chow for smooth algebraic groups), cf. [Ray70], VI 2.6. On the other hand, over the artinian local ring $A = \mathbf{C}[\varepsilon]/(\varepsilon^2)$ there exist abelian A -schemes G which are not projective (loc. cit., XII 4.2).

(37) **N.D.E.** What precedes holds for every base change $A \rightarrow A'$.

(38) **N.D.E.** This is expanded in 4.6 below: one must see that formation of the direct image under the morphisms p, λ, pr_2 commutes with flat base changes $A \rightarrow A'$. Since F and G are of finite type over artinian A , the morphisms f in question are all quasi-compact and quasi-separated, and the equality $f_*(\mathcal{O}_X) \otimes_A A' = f'_*(\mathcal{O}_{X'})$ (with obvious notation) follows from EGA IV₁, 1.7.21.

(39) **N.D.E.** Indeed, let $y_1, \dots, y_n$ be arbitrary points of X ; since X is of finite type over A , each y_i has in its closure a closed point x_i , and every open subset containing x_i contains y_i .

(40) **N.D.E.** Such an open subset exists, since X is of finite type over k : X has finitely many irreducible components $C_1, \dots, C_p$, and it suffices to take, for each i , a nonempty affine open subset contained in $C_i - \bigcup_{j \neq i} C_j$. (Here one further knows, by (ii'), that the C_i are disjoint...)

(41) **N.D.E.** The original has been expanded in what follows.

3.2.2. — For every algebra A' local, finite and free over A , denote now by $U(A')$ the set of points of $G \otimes_A A'$ having a saturated open neighborhood W such that the groupoid induced by $G_* \otimes_A A'$ on W has a quasi-section. It is quite clear that $U(A')$ is saturated for the actions of $G(\text{Spec } A')$ on $G \otimes_A A'$. We shall see that, when A' is sufficiently large, $U(A')$ equals $G \otimes_A A'$.

By Theorem V 8.1, $U(A)$ is nonempty, hence contains a closed point y . The proof then proceeds by induction on $\dim(G - U(A))$. Let $g_1, \dots, g_n$ be closed points belonging to the various irreducible components of $G - U(A)$. By 1.2, there exists A' local, finite and free over A such that the points $g'_1, \dots, g'_p$ (resp. $x = x_1, \dots, x_r$) of $G' = G \otimes_A A'$ projecting to $g_1, \dots, g_n$ (resp. y) are strictly rational over A' . Then $U(A')$ contains $(U(A) \otimes_A A') \cdot x^{-1}g'_i$ for every i ; hence $U(A')$ contains $g'_1, \dots, g'_p$, and one has

$$\dim(G' - U(A')) < \dim(G - U(A)).$$

The induction hypothesis then implies the existence of an algebra A'' local, finite and free over A' such that $U(A'') = G' \otimes_{A'} A'' = G \otimes_A A''$.

3.2.3. — We are now in a position to prove the existence of $F \backslash G$ when F and G are of finite type over A . Let A' be sufficiently large over A for $U(A')$ to coincide with $G \otimes_A A'$ (cf. 3.2.2). We shall put $A'' = A' \otimes_A A'$, and, for every A -scheme X , denote by X' and X'' the fibered products $X \otimes_A A'$ and $X \otimes_A A''$. By 3.2.1 and 3.2.2, the quotients $F' \backslash G'$ and $F'' \backslash G''$ exist, and one has the following commutative diagram, in which the first two rows and columns are exact:

$$(*) \quad \begin{array}{ccccc} F'' \times_{A''} G'' & \xrightarrow[\lambda'']{\text{pr}_2''} & G'' & \xrightarrow{p''} & F'' \backslash G'' \\ \downarrow w_2 \quad \downarrow w_1 & & \downarrow v_2 \quad \downarrow v_1 & & \downarrow u_2 \quad \downarrow u_1 \\ F' \times_{A'} G' & \xrightarrow[\lambda']{\text{pr}_2'} & G' & \xrightarrow{p'} & F' \backslash G' \\ \downarrow h & & \downarrow g & & \\ F \times_A G & \xrightarrow[\lambda]{\text{pr}_2} & G & & \end{array} .$$

In this diagram, pr_2' and λ' (resp. pr_2'' and λ'') are obtained from pr_2 and λ by obvious base changes; the morphisms g and h are induced by the canonical injection $A \rightarrow A'$. The canonical morphisms are denoted by p' and p'' ; the morphisms v_1, v_2 and w_1, w_2 are induced by the two canonical injections of A' into A'' . Finally, since construction of the quotient $F' \backslash G'$ commutes with the two base changes $f_1, f_2 : \text{Spec } A'' \rightrightarrows \text{Spec } A'$, one has, denoting by $\pi' : F' \backslash G' \rightarrow \text{Spec } A'$ the structure morphism, canonical isomorphisms for $i = 1, 2$:

$$\tau_i : F'' \backslash G'' \xrightarrow{\sim} (F' \backslash G') \times_{\pi', f_i} \text{Spec } A'',$$

and the morphism u_i is the composite of τ_i and the projection $(F' \backslash G') \times_{\pi', f_i} \text{Spec } A'' \rightarrow F' \backslash G'$.

Now, when one has a diagram of type $(*)$, the first two rows and columns being exact, one easily verifies that $\text{Coker}(\text{pr}_2, \lambda)$ exists if and only if $\text{Coker}(u_1, u_2)$ does, and these two cokernels are identified. The existence of $F \backslash G$ will therefore follow from that of $\text{Coker}(u_1, u_2)$.

Now it follows from compatibility of formation of $F \backslash G$ with the extensions of the base considered here (cf. N.D.E. (37) in 3.2.1, and 4.6 below) that the composite morphism

$$(F' \backslash G') \times_{\pi', f_1} \text{Spec } A'' \xrightarrow{\tau_1^{-1}} F'' \backslash G'' \xrightarrow{\tau_2} (F' \backslash G') \times_{\pi', f_2} \text{Spec } A''$$

is a descent datum on $F' \backslash G'$ relative to $f : \text{Spec } A' \rightarrow \text{Spec } A$. By 3.2.1 (†) and SGA 1, VIII 7.6, this descent datum is effective, that is, $\text{Coker}(u_1, u_2)$ exists (one could moreover use directly Theorem 4.1 of Exp. V).

3.2.4. — To finish the proof of assertions (i), (ii), (iii) and (iv)(a) of 3.2 in the case where F and G are of finite type over A , it remains to study the quotient $F \backslash G$. By V 6.1, assertions

(ii), (iii) and (iv)(a) “become true” after the base change $f : \operatorname{Spec} A' \rightarrow \operatorname{Spec} A$; by EGA IV₂, 2.6.1, 2.6.2 and 2.7.1, these assertions were therefore true before the base change. Finally, to prove the second assertion of (i), i.e. that $F \backslash G$ is the cokernel of $(\operatorname{pr}_2, \lambda)$ in the category of all ringed spaces, one need only refer to V §6.c).

3.2.5. — ⁽⁴²⁾Let us now show assertion (iv)(c) of 3.2, using the proof of VIB, 9.2.1. Denote by $X = F \backslash G$ and by d the morphism $F \times_A G \rightarrow G \times_A G$ with components λ and pr_2 .

Since u is a closed immersion by 2.5.2, and $d = \sigma \circ (u \times \operatorname{id}_G)$, where σ is the automorphism of $G \times_A G$ defined by $\sigma(x, y) = (xy, y)$, d is a closed immersion. On the other hand, by (iv)(a), one has the cartesian square below

$$\begin{array}{ccc} F \times_A G & \xrightarrow{d} & G \times_A G \\ \downarrow & & \downarrow p \times p \\ X & \xrightarrow{\Delta_X} & X \times_A X \end{array}$$

and p , hence also $p \times p$, is faithfully flat and locally of finite presentation. Thus, by fppf descent, since d is a closed immersion, so is Δ_X , i.e. X is separated.

3.3. — ⁽⁴³⁾In Theorem 3.2, the hypothesis that G is flat can be dropped when $u : F \rightarrow G$ is a monomorphism. This generalization is mentioned in Remark 9.3 b) of Exp. VIB, and also in [Ray67a], Example a) i), p. 82. The proof, found in Theorem 4 of [An73], follows from Theorem 3.2 and the following theorem of Grothendieck (mentioned in [Ray67a], Th. 1 ii) and proved in [DG70], §III.2, 7.1). If X is a scheme and R an equivalence relation on X , $\widetilde{X/R}$ will denote the quotient fppf sheaf of X by R (cf. IV 4.4.9).

Theorem 3.3.1 (Grothendieck). *Let A be a ring, X an A -scheme, and $d_0, d_1 : R \rightarrow X$ an A -equivalence relation on X such that d_i is faithfully flat and of finite presentation. Let X_0 be a saturated closed subscheme of X defined by a nilpotent ideal, and R_0 the equivalence relation induced by R on X_0 . Then, if the quotient fppf sheaf $\widetilde{X_0/R_0}$ is represented by an A -scheme, the same holds for $\widetilde{X/R}$.*

For the proof, we refer to [DG70], §III.2, 7.1. Now return to the case where A is an artinian local ring. Let $u : F \rightarrow G$ be a quasi-compact morphism between A -groups locally of finite type, and further suppose u a monomorphism. Then, by 2.5.2, u is a closed immersion.

One can now state the following variant of Theorem 3.2.

Theorem 3.3.2. *Let A be an artinian local ring, G an A -group locally of finite type, and F a closed subgroup of G , flat over A .⁽⁴⁴⁾ Then:*

- (i) *The quotient fppf sheaf $\widetilde{F \backslash G}$ is represented by a separated A -scheme $F \backslash G$ locally of finite type; moreover, the sequence*

$$F \times_A G \rightrightarrows G \xrightarrow{p} F \backslash G$$

is exact in the category of all ringed spaces.

- (ii) *$F \times_A G \xrightarrow{(\lambda, \operatorname{pr}_2)} G \times_{(F \backslash G)} G$ is an isomorphism, and $p : G \rightarrow F \backslash G$ is faithfully flat and locally of finite presentation, so p is an F -torsor locally trivial for the fppf topology.*
- (iii) *If G is flat (resp. of finite type, resp. smooth) over A , $F \backslash G$ is also so.*

⁽⁴²⁾**N.D.E.** This paragraph has been added.

⁽⁴³⁾**N.D.E.** This subsection has been added.

⁽⁴⁴⁾**N.D.E.** For an example where F is not flat and $\widetilde{F \backslash G}$ is not representable, see [DG70], §III.3, no. 3.3.

- (iv) $X = F \backslash G$ has a right action of G such that $p(e) \cdot g = p(g)$ for every $g \in G$; consequently the connected components of X are of finite type over A , irreducible and all of dimension $\dim G - \dim F$.
- (v) If moreover F is an invariant subgroup of G , there exists one and only one A -group structure on $F \backslash G$ such that $p : G \rightarrow F \backslash G$ is a morphism of A -groups.

Assertions (i) and (ii) follow from 3.2 and 3.3.1, and, since $G \rightarrow F \backslash G$ is faithfully flat and locally of finite presentation, assertion (iii) follows from EGA IV, 2.2.14, 2.7.1 and 17.7.7. We shall prove assertions (iv) and (v) in section 5. Let us immediately note the following corollary.

Corollary 3.3.3. *Let A be an artinian local ring, G an A -group locally of finite type, and H a closed subgroup of G , flat over A . Denote by p the morphism $G \rightarrow G/H$, and by λ (resp. pr_1) the morphism $G \times H \rightarrow G$ defined by $\lambda(g, h) = gh$ (resp. the projection $G \times H \rightarrow G$). Then, for every open subset U of G/H , one has*

$$\mathcal{O}(U) = \{\varphi \in \mathcal{O}(p^{-1}(U)) \mid \varphi \circ \lambda = \varphi \circ \text{pr}_1\},$$

i.e. $\mathcal{O}(U)$ is the set of $\varphi \in \mathcal{O}(p^{-1}(U))$ such that $\varphi(gh) = \varphi(g)$ for every A -scheme S and $g \in G(S)$, $h \in H(S)$.

Indeed, since $p : G \rightarrow G/H$ is faithfully flat and locally of finite presentation, hence a covering for the fppf topology, this follows from IV, 3.3.3.2.

4. Construction of quotients $F \backslash G$ (general case)

We now suppose the hypotheses of Theorem 3.2 satisfied, F and G not necessarily being of finite type over A .

4.1. — First consider a connected component G^α of G , and let us show that the saturation $\mathcal{S}(G^\alpha)^{(45)}$ of G^α for the equivalence relation defined by the groupoid G_* is an open and closed subset of G (in other words, the union of certain connected components of G).

This saturation is the image of $F \times_A G^\alpha$ under λ , hence is open in G (cf. §3.1). If k is the residue field of A and $\bar{k}$ an algebraic closure of k , it remains to show that the image of $(F \times_A G^\alpha) \otimes_A k$ under $\lambda \otimes_A k$ is closed in $G \otimes_A k$, or, by SGA 1, VIII.4.4, that the image of $(F \times_A G^\alpha) \otimes_A \bar{k}$ under $\lambda \otimes_A \bar{k}$ is closed. Since $G^\alpha \otimes_A \bar{k}$ is the union of finitely many connected components of $G \otimes_A \bar{k}$, one reduces to the case where A is an algebraically closed field, which we shall suppose. In this case, $\mathcal{S}(G^\alpha)$ is the union of the images of G^α under left translations $\ell_{u(x)}$, where x ranges over the closed points of F ; the assertion therefore follows from the fact that these images are connected components of G .

4.2. — Take, in particular, for G^α the connected component G^0 of the origin of G . Then $\mathcal{S}(G^0)$ obviously contains the image of F under u , which is none other than the equivalence class of the origin. On the other hand, if F^β is a connected component of F , $F^\beta \times_A G^0$ is connected (2.1.2), so the image of $F^\beta \times_A G^0$ under λ is contained in the connected component of $u(F^\beta)$ in G . In other words, $\mathcal{S}(G^0)$ is the union of the connected components meeting the image of F .

One will also note that the open subscheme of G whose underlying space is $\mathcal{S}(G^0)$ is a subgroup of G (which we again denote by $\mathcal{S}(G^0)$): indeed, the inversion morphism of G preserves the image of F and permutes the connected components of G meeting this image; it therefore suffices to show that $\nu : G \times_A G \rightarrow G$ sends $\mathcal{S}(G^0) \times_A \mathcal{S}(G^0)$ into $\mathcal{S}(G^0)$, and for this one can suppose A an algebraically closed field (with the notation of 4.1, $\mathcal{S}(G^0) \otimes_A \bar{k}$ is indeed identified with the saturation of $(G \otimes_A \bar{k})^0$ for the equivalence relation defined by the homomorphism $u \otimes_A \bar{k}$); if G^γ and G^δ are then connected components of $\mathcal{S}(G^0)$, $G^\gamma \times_A G^\delta$

⁽⁴⁵⁾**N.D.E.** The saturation of G^α has been denoted by $\mathcal{S}(G^\alpha)$ instead of $\overline{G}^\alpha$.

is connected and its image under ν meets the image of F ; consequently, $u(G^\gamma \times_A G^\delta)$ is contained in a connected component of G meeting $u(F)$.

4.3. — It follows from what precedes that the groupoid G_* with base G defined by u is the direct sum of the groupoids $\mathcal{S}(G_*^\alpha)$ induced by G_* on the various open and closed subsets of G of the form $\mathcal{S}(G^\alpha)$. The cokernel of G_* is therefore the direct sum of the cokernels of these groupoids $\mathcal{S}(G_*^\alpha)$, which one is led to study separately.

First consider the groupoid $\mathcal{S}(G_*^0)$ induced by G_* on $\mathcal{S}(G^0)$. It is clear that $\mathcal{S}(G_*^0)$ is the groupoid with base $\mathcal{S}(G^0)$ defined by the homomorphism from F to $\mathcal{S}(G^0)$ induced by u (§3.1). The cokernel whose existence we wish to prove is therefore identified with $F \setminus \mathcal{S}(G^0)$. Consider, on the other hand, the groupoid

$$G_2^0 \begin{smallmatrix} \xrightarrow{\ell_2'} \\ \xrightarrow{\ell_1'} \\ \xrightarrow{\ell_0'} \end{smallmatrix} G_1^0 \begin{smallmatrix} \xrightarrow{\ell_1} \\ \xrightarrow{\ell_0} \end{smallmatrix} G_0^0 = G^0$$

induced by $\mathcal{S}(G_*^0)$ on G^0 . If one refers to the construction made explicit in V §3.b), the object then denoted $Y_0 \times_{X_0} X_1$ is none other than $F \times_A G^0$, so G_1^0 is the inverse image of G^0 under the morphism $F \times_A G^0 \rightarrow \mathcal{S}(G^0)$ induced by λ .

I say that this inverse image is $F_0 \times_A G^0$, where F_0 denotes the inverse image of G^0 under u . Indeed, if F^β is a connected component of F_0 , $F^\beta \times_A G^0$ is connected (2.1.2), and $\lambda(F \times_A G^0)$ is contained in G^0 ; conversely, if F^β is a connected component of F not contained in F_0 , the image of $F^\beta \times_A G^0$ is again connected and contains $u(F^\beta)$; if $u(F^\beta)$ is not contained in G^0 , $\lambda(F^\beta \times_A G^0)$ does not meet G^0 .

It follows from what precedes that the groupoid G_*^0 induced by G_* on G^0 is the groupoid with base G^0 defined by the homomorphism $F_0 \rightarrow G^0$ induced by u . Since G^0 , and hence F_0 , are of finite type over A , by paragraph 4, G_*^0 has a cokernel which is none other than $F_0 \setminus G^0$.

I now say that $F_0 \setminus G^0$ is identified with $F \setminus \mathcal{S}(G^0)$. Indeed, the proof is analogous to that of the first part of assertion (i) of Lemma V §6.1; consider the diagram:

$$\mathcal{S}(G^0) \xleftarrow{v} F \times_A G^0 \xrightarrow{\text{pr}_2} G^0,$$

where v is the morphism induced by λ . Since pr_2 has a section, pr_2 is an effective universal epimorphism, so $F_0 \setminus G^0$ coincides with $\text{Coker}(v_0, v_1)$, where

$$V_2 \begin{smallmatrix} \xrightarrow{v_2'} \\ \xrightarrow{v_1'} \\ \xrightarrow{v_0'} \end{smallmatrix} V_1 \begin{smallmatrix} \xrightarrow{v_1} \\ \xrightarrow{v_0} \end{smallmatrix} V = F \times_A G^0$$

is the inverse image under pr_2 of the groupoid G_*^0 (cf. V §3.a), that is, also the inverse image of $\mathcal{S}(G_*^0)$ under the composite morphism

$$F \times_A G^0 \xrightarrow{\text{inclusion}} F \times_A \mathcal{S}(G^0) \xrightarrow{\text{pr}_2} \mathcal{S}(G^0).$$

Similarly, since v is faithfully flat and quasi-compact, $F \setminus \mathcal{S}(G^0)$ coincides with the cokernel of the inverse image of $\mathcal{S}(G_*^0)$ under the base change v . Now this inverse image is isomorphic to V_* by Exp. V, §3.c; it follows that the canonical inclusion of G_*^0 into $\mathcal{S}(G_*^0)$ induces an isomorphism from $F_0 \setminus G^0$ onto $F \setminus \mathcal{S}(G^0)$.

Finally, note that construction of $F \setminus \mathcal{S}(G^0)$ commutes with finite locally free base changes, since the same holds for $F_0 \setminus G^0$ (cf. N.D.E. (37) and 4.6 below).

4.4. — It remains to construct the cokernel of the groupoid $\mathcal{S}(G_*^\alpha)$ when G^α is any connected component of G . If A' is a sufficiently large local A -algebra finite and free (cf. 3.2), $G^\alpha \otimes_A A'$ is the union of finitely many connected components $C_1, \dots, C_n$ of $G \otimes_A A'$, all having a strictly rational point. For every i , there therefore exists a right translation r_i of $G \otimes_A A'$ sending

$G^0 \otimes_A A'$ onto C_i ; this translation induces an isomorphism from the groupoid $\mathcal{S}(G_*^0) \otimes_A A'$ onto $\mathcal{S}(C_*^i)$, so the groupoid induced by $G_* \otimes_A A'$ on the saturation of C_i has a cokernel.

Since $\mathcal{S}(G_*^\alpha) \otimes_A A'$ is the direct sum of a certain number of the $\mathcal{S}(C_*^i)$, $\mathcal{S}(G_*^\alpha) \otimes_A A'$ has a cokernel; this cokernel is the direct sum of a certain number of copies of $(F_0 \otimes_A A') \setminus (G^0 \otimes_A A')$, so every finite subset of this cokernel is contained in an affine open subset; moreover, construction of this cokernel commutes with finite locally free extensions of the base (cf. N.D.E. (37) and 4.6 below). One therefore sees, as in 3.2.3, that this cokernel has the form $Y \otimes_A A'$, where Y is a cokernel of $\mathcal{S}(G_*^\alpha)$.

4.5. — We have therefore constructed $F \setminus G$ and shown that it is a direct sum of schemes of finite type over A . The other assertions of Theorem 3.2 reduce directly to assertions concerning the groupoids $\mathcal{S}(G_*^\alpha)$. As in V §6, the second assertion of (i) follows from the first and from (ii) and (iii), so it suffices to prove (ii), (iii) and (iv)(a). Since A' is a local A -algebra finite and free, the morphism $A \rightarrow A'$ is faithfully flat and of finite presentation, so, by SGA 1, VIII (3.1, 4.6, 5.4), it suffices to verify the corresponding assertions for the groupoid $\mathcal{S}(G_*^\alpha) \otimes_A A'$. Now the latter is isomorphic to the direct sum of finitely many copies of $\mathcal{S}(G_*^0) \otimes_A A'$ (cf. 4.4), so one is reduced to the groupoid $\mathcal{S}(G_*^0)$.

For the latter one continues to follow the proof established in V §6, as we began to do in 4.3.

4.6. — To finish this paragraph, let us add some remarks concerning Lemma 6.1 and §9.a of Exposé V: with the hypotheses and notation of V §9.a, we seek a condition under which construction of the cokernel of the (Sch/S) -groupoid X_* commutes with an extension $\pi : S' \rightarrow S$ of the base. Since the cokernels of X_* and X'_* are identified with the cokernels of the groupoids U_* and U'_* induced by X_* and X'_* on the quasi-sections U and U' , one is reduced to the case of a (Sch/S) -groupoid satisfying the hypotheses of Theorem V 4.1.

⁽⁴⁶⁾If Y denotes the cokernel of U_* , $Y' = Y \times_S S'$, and Y_1 the cokernel of U'_* , we have seen in V §9.a that the canonical morphism $Y_1 \rightarrow Y'$ is a homeomorphism (and even a universal homeomorphism); one can therefore identify Y_1 and Y' as topological spaces. If $p : U \rightarrow Y$ is the canonical morphism and $p' : U' \rightarrow Y'$ is obtained from it by base change, we then want the sequence of $\mathcal{O}_{Y'}$ -modules

$$(*) \quad \mathcal{O}_{Y'} \longrightarrow p'_*(\mathcal{O}_{U'}) \rightrightarrows p'_*u'_{1*}(\mathcal{O}_{U'_1}) = p'_*u'_{0*}(\mathcal{O}_{U'_1})$$

to be exact.⁽⁴⁷⁾ Since we have placed ourselves under the hypotheses of V 4.1, u_0 and u_1 are finite locally free; and, by V.4.1 (ii), p is integral. Then p and $p \circ u_i$ are affine, hence separated and quasi-compact.

Consequently, if S' is flat over S , it follows from EGA III₁, 1.4.15 (taking account of correction ErrIII 25 in EGA III₂) that sequence $(*)$ is identified with the inverse image of the sequence

$$(**) \quad \mathcal{O}_Y \longrightarrow p_*(\mathcal{O}_U) \rightrightarrows p_*u_{1*}(\mathcal{O}_{U_1}) = p_*u_{0*}(\mathcal{O}_{U_1}),$$

which is an exact sequence.⁽⁴⁸⁾ An analogous argument applies when the groupoid X_* “locally” has quasi-sections (cf. the proof of Theorem V 7.1). One therefore obtains:

Proposition 4.6.1. *Construction of the cokernel of X_* commutes with flat extensions of the base when X_* locally has quasi-sections.*

⁽⁴⁶⁾N.D.E. The following sentence has been added.

⁽⁴⁷⁾N.D.E. In what follows, the original has been modified, the additional hypotheses made on X_* being superfluous.

⁽⁴⁸⁾N.D.E. The following sentence and the numbering 4.6.1 below have been added to bring out the stated result.

4.7. — Now consider the case of the groupoid G_* of Theorem 3.2, temporarily supposing F and G of finite type over A .

By 3.2.2, there exists an algebra A' local, finite and free over A such that the groupoid $G_* \otimes_A A'$ “locally” has quasi-sections. For every extension $T \rightarrow \operatorname{Spec} A$ of the base, the sequence

$$(F'' \backslash G'') \times_{\operatorname{Spec} A} T \rightrightarrows (F' \backslash G') \times_{\operatorname{Spec} A} T \longrightarrow (F \backslash G) \times_{\operatorname{Spec} A} T$$

obtained from diagram (*) of 3.2.3 is exact. If one further supposes T flat over $\operatorname{Spec} A$, $(F'' \backslash G'') \times_{\operatorname{Spec} A} T$ and $(F' \backslash G') \times_{\operatorname{Spec} A} T$ are identified respectively, by 4.6, with the cokernels of the groupoids

$$(G_* \otimes_A A'') \times_{\operatorname{Spec} A} T \quad \text{and} \quad (G_* \otimes_A A') \times_{\operatorname{Spec} A} T.$$

The diagram obtained from 3.2.3 (*) by the base change $T \rightarrow \operatorname{Spec} A$ then shows that $(F \backslash G) \times_{\operatorname{Spec} A} T$ is identified with the cokernel of $G_* \times_{\operatorname{Spec} A} T$. An analogous argument holds in the general case (i.e. when G and F are locally of finite type over A). One therefore obtains:⁽⁴⁹⁾

Proposition 4.7.1. *Under the hypotheses of Theorem 3.2, for every flat A -scheme T , $(F \backslash G) \times_{\operatorname{Spec} A} T$ is identified with the left quotient of $G \times_{\operatorname{Spec} A} T$ by $F \times_{\operatorname{Spec} A} T$.*

5. Connections with Exposé IV and consequences

5.1. — ⁽⁵⁰⁾We return to the notation of §3 and the hypotheses of Theorem 3.2; one then has the following commutative diagram

$$\begin{array}{ccc} F \times_A G \times_A G & \xrightarrow{F \times \nu} & F \times_A G \\ \lambda \times G \downarrow \text{pr}_2 \times G & & \lambda \downarrow \text{pr}_2 \\ G \times_A G & \xrightarrow{\nu} & G \\ p \times G \downarrow & & \downarrow p \\ (F \backslash G) \times_A G & \xrightarrow[\rho]{} & F \backslash G, \end{array}$$

which satisfies $\text{pr}_2 \circ (F \times \nu) = \nu \circ (\text{pr}_2 \times G)$ and $\lambda \circ (F \times \nu) = \nu \circ (\lambda \times G)$. Moreover, since G is supposed flat over A , the vertical sequence on the left is exact by 4.7, so ν induces a morphism of A -schemes:

$$\rho : (F \backslash G) \times_A G \longrightarrow F \backslash G.$$

This morphism ρ makes G act on $F \backslash G$ on the right, as one verifies immediately; moreover, the canonical morphism $G \rightarrow F \backslash G$ commutes with the right actions of G on G and $F \backslash G$.

⁽⁵¹⁾This proves the first assertion of point (ii') of 3.2. By 2.5.4, one then obtains that the connected components of $X = F \backslash G$ are of finite type, irreducible and all of the same dimension. To evaluate this dimension, one can suppose $A = k$ and k algebraically closed. By I, 2.3.3.1, the stabilizer of the k -point $p(e)$ is represented by the fiber $H = p^{-1}(p(e))$, and, since $F \backslash G$ is the quotient of G by F in the category of ringed spaces, this fiber has underlying space $u(F)$; since $\operatorname{Ker}(u)$ is finite, one therefore has $\dim H = \dim u(F) = \dim F$. By 2.5.4 (ii), one therefore obtains $\dim X = \dim G - \dim F$. This proves point (ii') of Theorem 3.2 (and hence also point (iv) of 3.3.2).

5.2. — When the homomorphism of A -groups $u : F \rightarrow G$ is a monomorphism, one can recover 5.1 by using the results of Exposé IV. Indeed, the canonical morphism $p : G \rightarrow F \backslash G$

⁽⁴⁹⁾**N.D.E.** The numbering 4.7.1 below has been added to bring out the stated result.

⁽⁵⁰⁾**N.D.E.** The title of this section (called “Supplements” in the original) has been changed.

⁽⁵¹⁾**N.D.E.** What follows has been added.

is faithfully flat and open by 3.2; it is therefore covering for the fpqc topology (IV 6.3.1), and one can apply Corollaries IV.5.2.2 and IV.5.2.4.

In particular, if we suppose, in addition to the hypotheses of 3.2, that u is the inclusion of an invariant subgroup F into G , there exists one and only one A -group structure on $F \backslash G$ such that the canonical morphism $p : G \rightarrow F \backslash G$ is a homomorphism of A -groups.⁽⁵²⁾ This proves point (v) of 3.3.2.

5.3. — We shall now review some statements of Exposé IV.

5.3.1. — Statements IV 5.2.7 and IV 5.3.1 translate as follows. Let F and G be two groups locally of finite type and flat over A , F being an invariant closed subgroup of G . The maps $H \mapsto F \backslash H$ and $H' \mapsto H' \times_{(F \backslash G)} G$ define a bijective correspondence between flat A -subgroups of G containing F and flat A -subgroups of $F \backslash G$. In this bijection, closed (resp. invariant) subgroups of G containing F correspond to closed (resp. invariant) subgroups of $F \backslash G$.⁽⁵³⁾

5.3.2. — Proposition IV 5.2.9 implies the following result. Let F, H, G be groups locally of finite type and flat over A ; suppose $F \subset H \subset G$, with F closed in G and invariant in H . Under these conditions, $F \backslash H$ acts freely on $F \backslash G$ on the left, the quotient scheme $(F \backslash H) \backslash (F \backslash G)$ exists, and one has a canonical isomorphism of schemes with operator group G :

$$(F \backslash H) \backslash (F \backslash G) = H \backslash G.$$

5.3.3. — Finally, from IV 5.2.8 follows the assertion below. Let F, H, G be groups locally of finite type and flat over A ; suppose F contained, closed and invariant in G , H contained in G , and $F \cap H$ flat over A . Let, then, $F \times_A^\tau H$ be the A -group with underlying scheme the product $F \times_A H$, multiplication being defined by the morphism “ $((x, h), (y, h')) \mapsto (xhyh^{-1}, hh')$ ”; similarly, let $u : H \cap F \rightarrow F \times_A^\tau H$ be the monomorphism $x \mapsto (x^{-1}, x)$, and $F \cdot H$ the quotient $(F \cap H) \backslash (F \times_A^\tau H)$. Under these conditions, there exists a canonical isomorphism

$$F \backslash (F \cdot H) = (F \cap H) \backslash H.$$

5.4. —⁽⁵⁴⁾ Let $u : G \rightarrow H$ be a quasi-compact morphism between A -groups locally of finite type, whose kernel N is flat over A . In this case, by 3.3.2 and 5.2, the quotient A -group $C = N \backslash G$ exists, and the morphism $p : G \rightarrow C$ is faithfully flat and locally of finite presentation. On the other hand, by IV 5.2.6, u induces a monomorphism $v : C \rightarrow H$, which is quasi-compact (since u is and $G \rightarrow C$ is surjective, cf. EGA IV₁, 1.1.3), hence is a closed immersion by 2.5.2. One has therefore obtained the following proposition:

Proposition 5.4.1. *Let $u : G \rightarrow H$ be a quasi-compact morphism between A -groups locally of finite type, such that $N = \text{Ker } u$ is flat over A . Then one has the factorization:*

$$\begin{array}{ccc} G & \xrightarrow{u} & H \\ p \downarrow & \nearrow i & \\ N \backslash G & & \end{array}$$

where p is faithfully flat and locally of finite presentation, and i a closed immersion.

Further suppose G flat over A . Then, by 3.3.2, $C = N \backslash G$ is flat over A , and hence the quotient $X = C \backslash H$ exists in (Sch/A) and represents the quotient fppf sheaf $\widetilde{C \backslash H}$, and

⁽⁵²⁾**N.D.E.** The following sentence has been added.

⁽⁵³⁾**N.D.E.** In addition to the cited statements of Exp. IV, one uses the fact that, since $G \rightarrow F \backslash G$ is faithfully flat, an A -subgroup H of G is flat over A if and only if $F \backslash H$ is.

⁽⁵⁴⁾**N.D.E.** The original has been expanded in what follows; in particular, Proposition 5.4.1 has been added.

$q : H \rightarrow X$ is a C -torsor. Consequently, denoting by $e : \operatorname{Spec} A \rightarrow G$ the unit section of G , v induces an isomorphism of fppf sheaves between $\tilde{C}$ and the fibered product of q and $q \circ e : \operatorname{Spec} A \rightarrow X$, represented by a closed subscheme of H . Consequently, v is an isomorphism of C onto a closed subgroup scheme K of G (equal to the stabilizer of the A -point $q \circ e$ of X). (This provides another proof that every quasi-compact monomorphism $v : C \rightarrow H$ between A -groups locally of finite type is a closed immersion, cf. 2.5.2 and VIB 1.4.2.)

Further suppose C an invariant subgroup of H ; in this case, the A -group $\overline{H} = C \backslash H$ is the cokernel in the category of A -groups of the morphism $u : G \rightarrow H$, and K is the kernel of the morphism $H \rightarrow \overline{H}$. When G and H are abelian A -groups, K is the image of u in the category of abelian A -groups, whereas $C = (\operatorname{Ker} u) \backslash G$ is the coimage of u . Taking account of the isomorphism $C \xrightarrow{\sim} K$ just established, one obtains:⁽⁵⁵⁾

Theorem 5.4.2. *Let k be a field. The category of commutative algebraic k -groups is abelian.*

Indeed, when k is a field, $\operatorname{Ker}(u)$ is flat over k whatever u may be.

⁽⁵⁶⁾Note that the full subcategory of affine commutative algebraic k -groups is thick. Indeed, consider an exact sequence of commutative algebraic k -groups:

$$1 \longrightarrow N \longrightarrow G \longrightarrow G/N \longrightarrow 1.$$

If G is affine, it is clear that N is so, and G/N is also so by a theorem of Chevalley, cf. VIB, 11.17. Conversely, if N and G/N are affine, G is also so by VIB, 9.2 (viii). One therefore obtains:

Corollary 5.4.3. *Let k be a field. The category of affine commutative algebraic k -groups is abelian.*

Let us further mention that the category of all affine commutative k -groups (not necessarily of finite type) is abelian; this follows from VIB, 11.17 and 11.18.2 (cf. [DG70], §III.3, 7.4); see also VIIB, 2.4.2 for a proof using formal groups.

5.5. — Let G be a group locally of finite type and flat over an artinian local ring A . One knows (2.3) that the connected component G^0 of the origin is an invariant open subgroup scheme of G , hence also flat over A . Then, by 3.2 and 5.2, $G^0 \backslash G$ is an A -group scheme flat over A . Moreover, since each connected component G^α of G is saturated for the equivalence relation defined by G^0 , $G^0 \backslash G$ is the direct sum of the $G^0 \backslash G^\alpha$ (cf. 4.3). In particular, the connected component of the origin in $G^0 \backslash G$ is none other than $G^0 \backslash G^0 \simeq \operatorname{Spec} A$, and hence $G^0 \backslash G \rightarrow \operatorname{Spec} A$ is a local isomorphism at the origin. Consequently, $G^0 \backslash G$ is étale over $\operatorname{Spec} A$, by VIB, 1.3.⁽⁵⁷⁾ One therefore obtains the following proposition (for point (ii), we refer to [DG70], §II.5, 1.7–1.10):

Proposition 5.5.1. *Let A be an artinian local ring and G an A -group locally of finite type and flat.*

- (i) $G^0 \backslash G$ is an étale A -group.
- (ii) Consequently, if $A = k$ is an algebraically closed field, $G^0 \backslash G$ is a constant k -group, acting simply transitively on the set of connected components of G ; hence, if G is algebraic, $G^0 \backslash G$ is finite.

5.6. — Let now k be a perfect field and G a k -group locally of finite type. We have seen (0.2) that G_{red} is then a subgroup scheme of G . Moreover, the equivalence class of the origin

⁽⁵⁵⁾**N.D.E.** The number 5.4.2 has been added to this theorem.

⁽⁵⁶⁾**N.D.E.** What follows has been added.

⁽⁵⁷⁾**N.D.E.** The original has been expanded in what precedes, and Proposition 5.5.1 has been added to bring out this result.

of G for the left action of G_{red} on G is the whole underlying space of G . Thus, by Theorem 3.2, one obtains:

Proposition 5.6.1. *Let k be a perfect field and G a k -group locally of finite type. Then the k -scheme $G_{\text{red}} \backslash G$ is the spectrum of a finite local k -algebra with residue field k .*

⁽⁵⁸⁾Indeed, by 3.2, $G_{\text{red}} \backslash G$ has only one point, with residue field k , and is a k -scheme of finite type; it is therefore the spectrum of a finite-dimensional local k -algebra (cf. EGA I, 6.4.4).

Proposition 5.6.2. *Let $u : F \rightarrow G$ be a morphism between groups locally of finite type over a perfect field k . The following assertions are equivalent:*

- (i) u is flat.
- (ii) $u^0 : F^0 \rightarrow G^0$ is dominant, and the morphism $v : F_{\text{red}} \backslash F \rightarrow G_{\text{red}} \backslash G$ induced by u is flat.

⁽⁵⁹⁾Indeed, consider the following commutative diagram:

$$\begin{array}{ccc} F & \xrightarrow{p} & F_{\text{red}} \backslash F \\ u \downarrow & & \downarrow v \\ G & \xrightarrow{q} & G_{\text{red}} \backslash G, \end{array}$$

where p and q denote the canonical projections. By 3.2 (iv), p and q are faithfully flat; consequently, if u is flat, $q \circ u = v \circ p$ is flat, and therefore so is v .

Conversely, suppose v flat and u^0 dominant. Since u^0 is quasi-compact (F^0 being of finite type over k by 2.4, hence noetherian), it sends the generic point ξ of F^0 to the generic point η of G^0 . Let R be the finite local k -algebra whose spectrum is $G_{\text{red}} \backslash G$, and $\mathfrak{m}$ its maximal ideal. One has local morphisms of local rings $R \rightarrow \mathcal{O}_{G,\eta} \rightarrow \mathcal{O}_{F,\xi}$. Note that one has a cartesian square:

$$\begin{array}{ccc} G_{\text{red}} & \longrightarrow & G \\ \downarrow & & \downarrow q \\ \text{Spec}(R/\mathfrak{m}) & \longrightarrow & \text{Spec}(R) \end{array}$$

and therefore $\mathcal{O}_{G,\eta}/\mathfrak{m}\mathcal{O}_{G,\eta} \simeq \mathcal{O}_{G_{\text{red}},\eta} = \kappa(\eta)$, so $\mathcal{O}_{F,\xi}/\mathfrak{m}\mathcal{O}_{F,\xi}$ is flat over $\mathcal{O}_{G,\eta}/\mathfrak{m}\mathcal{O}_{G,\eta}$.

On the other hand, since q and $v \circ p$ are flat, G and F are flat over R . Consequently, by the local flatness criterion (cf. EGA IV₃, 11.3.10.2), $\mathcal{O}_{F,\xi}$ is flat over $\mathcal{O}_{G,\eta}$, that is, u is flat at ξ . Thus, by 2.5.3, u is flat.

6. Supplements on k -groups not necessarily of finite type

⁽⁶⁰⁾Let us further mention the following results, which will be useful in the addition VIB, §12. Fix a base field k .

Lemma 6.1. *Let G be a k -group. For every $x \in G$, there exists a point $u \in G \times G$ such that $\mu(u) = x$ and both projections $p_1(u)$ and $p_2(u)$ are maximal points of G .*

PROOF. Put $K = \kappa(x)$. Since the projection $G_K \rightarrow G$ sends maximal points to maximal points, one reduces to the case where x is rational. Then left translation λ_x (resp. right translation ρ_x) gives a morphism $G \rightarrow G \times G$, $g \mapsto (\lambda_x(g^{-1}), g)$ (resp. $g \mapsto (g, \rho_x(g^{-1}))$),

⁽⁵⁸⁾**N.D.E.** The numbering 5.6.1 and the following proof have been added.

⁽⁵⁹⁾**N.D.E.** The proof of (ii) $\Rightarrow$ (i) has been expanded, and the diagram below simplified.

⁽⁶⁰⁾**N.D.E.** The following results, taken from [Per75], have been added. Note that Lemma 6.1 can be expressed, in Weil's language, by saying that "every point of G is a product of two generic points".

inducing an isomorphism from G onto $\mu^{-1}(x)$, inverse to p_2 (resp. p_1). Thus, if u is a maximal point of $\mu^{-1}(x)$, $p_1(u)$ and $p_2(u)$ are maximal points of G , so u has the required property. $\square$

Corollary 6.2. *Let $f : G \rightarrow H$ be a quasi-compact dominant morphism of k -groups.*

- (i) *f is surjective.*
- (ii) *If H is reduced, f is faithfully flat.*

PROOF. Denote by μ_H (resp. μ_G) multiplication in H (resp. G). Let $h \in H$. By 6.1, there exists $u \in H \times H$ such that $\mu_H(u) = h$ and $\alpha = p_1(u)$ and $\beta = p_2(u)$ are maximal points of H . Since f is quasi-compact and dominant, $f^{-1}(\alpha)$ and $f^{-1}(\beta)$ are nonempty (cf. EGA IV₁, 1.1.5), so there exists $v \in G \times G$ such that $(f \times f)(v) = u$ (cf. EGA I, 3.5.2). Then $g = \mu_G(v)$ satisfies $f(g) = h$. This shows that f is surjective.

Further suppose H reduced. Then $\mathcal{O}_{H,\alpha}$ is a field, and we have seen above that $f^{-1}(\alpha) \neq \emptyset$, so f is flat at every point ξ of $f^{-1}(\alpha)$, hence f is flat by Lemma 2.5.3. $\square$

Recall 6.3. Recall (cf. EGA IV₃, 11.10.1) that a morphism of schemes $f : X \rightarrow Y$ is said to be schematically dominant if it satisfies the following condition: for every open subset U of Y , if Z is a closed subscheme of U such that the morphism $f^{-1}(U) \rightarrow U$ factors through Z , then $Z = U$. When f is quasi-compact and quasi-separated, this amounts to saying that the closed image of X under f is Y (cf. loc. cit., 11.10.3 (iv) and EGA I, 9.5.8).

Proposition 6.4. *Let $f : H \rightarrow G$ be a quasi-compact morphism of k -groups. Then the closed image of f is a subgroup scheme H' of G , and f factors as:*

$$\begin{array}{ccc} H & \xrightarrow{f} & G \\ f' \downarrow & \nearrow i & \\ H' & & \end{array}$$

where f' is schematically dominant, quasi-compact and surjective.

PROOF. Since H is separated (0.3), f is quasi-compact and separated, so $f_*(\mathcal{O}_H)$ is a quasi-coherent $\mathcal{O}_G$ -module, and the closed image H' of f exists and is the closed subscheme of G defined by the quasi-coherent ideal $I = \text{Ker}(\mathcal{O}_G \rightarrow f_*(\mathcal{O}_H))$ (cf. EGA I, §9.5).

Denote by c_G, μ_G (resp. c_H, μ_H) the inversion and multiplication morphisms of G (resp. H). Then $c_G \circ f = f \circ c_H$ factors through the closed subscheme $c_G(H')$, whence $H' \subset c_G(H')$, and therefore $H' = c_G(H')$ (since $c_G^2 = \text{id}_G$). Similarly, since $f \circ \pi_H = \pi_G \circ (f \times f)$ factors through H' , $f \times f$ factors through the closed subscheme $\pi_G^{-1}(H')$ of $G \times G$. On the other hand, since formation of the closed image commutes with flat base changes (EGA III 1.4.15 and IV₁ 1.7.21), the closed image of $f \times \text{id}_H$ (resp. $\text{id}_{H'} \times f$) is $H' \times H$ (resp. $H' \times H'$). Thus, by “transitivity of closed images” (EGA I, 9.5.5), the closed image of $f \times f$ is $H' \times H'$, hence contained in $\pi_G^{-1}(H')$, i.e. the restriction of π_G to $H' \times H'$ factors through H' . This shows that H' is a closed subgroup scheme of G . Denote the inclusion $H' \hookrightarrow G$ by i .

Then $f = i \circ f'$, where $f' : H \rightarrow H'$ is schematically dominant and quasi-compact (since f is quasi-compact and i separated). Thus, by 6.2, f' is surjective. This proves 6.4. $\square$

One can now state the following theorem ([Per75], V 3.1 & 3.2; see also [Per76], 0.0 & 0.1).

Theorem 6.5 (D. Perrin). *Let G be a quasi-compact k -group. Then:*

- (i) *G is the projective limit of a filtered system (G_i) of k -groups of finite type (whose transition morphisms $u_{ij} : G_j \rightarrow G_i$ are affine for sufficiently large i), and the morphisms $G \rightarrow G_i$ are faithfully flat (and affine for sufficiently large i).*
- (ii) *Let H be a closed k -subgroup of G . Then the quotient fpqc sheaf $\widetilde{G/H}$ is a k -scheme in the following two cases:*

- (1) *The immersion $H \rightarrow G$ is of finite presentation; in this case, G/H is of finite type over k .*
- (2) *H is invariant in G .*

For the proof of this theorem (which rests on several intermediate theorems), we refer to [Per75]. For the reader's convenience, let us nevertheless prove the two corollaries below, cf. [Per75] V 3.3 to 3.4 or [Per76] 4.2.3 to 4.2.5.

Corollary 6.6. *Let $f : G \rightarrow H$ be a quasi-compact morphism of k -groups.*

- (i) *If f is schematically dominant, it is faithfully flat.*
- (ii) *This is the case, in particular, if H is affine and the morphism $f^\natural : \mathcal{O}(H) \rightarrow \mathcal{O}(G)$ is injective.*

PROOF. Suppose f schematically dominant. Then, by 6.2 (i), f is surjective, so, by 2.5.3, it suffices to show f flat at the neutral element of G . Since $\mathcal{O}_{H,e} = \mathcal{O}_{H^0,e}$ (cf. 2.6.5), one can replace H by H^0 , hence suppose H irreducible. Then H is quasi-compact (loc. cit.), and, since f is quasi-compact, so is G . By 6.5 (i), $H = \varprojlim_i H_i$, where each H_i is an algebraic k -group. Denote by N_i the kernel of $G \rightarrow H_i$; it is a closed invariant k -subgroup of G . Moreover, since the unit section $\{e\} \rightarrow H_i$ is of finite presentation, so is the immersion $N_i \rightarrow G$; hence, by 6.5 (ii), the fpqc quotient G/N_i is an algebraic k -group. One then has a commutative diagram

$$\begin{array}{ccc} G & \xrightarrow{f} & H \\ q_i \downarrow & & \downarrow p_i \\ G/N_i & \xrightarrow{f_i} & H_i \end{array}$$

where f is schematically dominant and p_i, q_i faithfully flat (hence schematically dominant). Then $f_i \circ q_i = p_i \circ f$ is schematically dominant, and hence so is f_i . On the other hand, f_i is a monomorphism, hence a closed immersion, since G/N_i and H_i are algebraic (2.5.2). It follows that f_i is an isomorphism, and therefore $G \rightarrow H_i$ is faithfully flat. Then, by [BAC] I §2.7, Prop. 9, the morphism $G \rightarrow H$ is flat; on the other hand, it is surjective by 6.2 (i), hence faithfully flat. This proves point (i), and point (ii) follows, for if H is affine and $f^\natural$ injective, the closed image of f equals H , so f is schematically dominant (cf. 6.4). $\square$

Corollary 6.7. *Let $u : G \rightarrow H$ be a morphism of k -groups, and $N = \text{Ker}(u)$. Suppose u quasi-compact.*

- (i) *The quotient fpqc sheaf $\widetilde{G/N}$ is represented by a k -group scheme G/N , and u factors as:*

$$\begin{array}{ccc} G & \xrightarrow{u} & H \\ p \downarrow & \nearrow i & \\ G/N & & \end{array}$$

where p is faithfully flat and i a closed immersion.

- (ii) *In particular, if u is a monomorphism, u is a closed immersion, and if u is schematically dominant, u is faithfully flat.*

PROOF. (i) By 6.4 and 6.6, the closed image G' of u is a closed subgroup scheme of G , and $p : G \rightarrow G'$ is faithfully flat and quasi-compact. One obviously has $\text{Ker}(p) = \text{Ker}(u) = N$, and hence, by Exp. IV, 3.3.2.1 and 5.1.7, G' represents the quotient fpqc sheaf $\widetilde{G/N}$.

(ii) The second assertion is contained in 6.6; let us show the first. If u is a monomorphism, so is p ; then p is both a monomorphism and an effective epimorphism, hence an isomorphism (cf. IV, 1.14). This proves 6.7. $\square$

Finally, let us mention the following corollaries (cf. [Per76], 4.2.6 to 4.2.8).

Corollary 6.8. *The category of quasi-compact commutative k -group schemes is abelian.*

Taking account of 6.7, the proof is analogous to that of 5.4.2.

Corollary 6.9. *If $\text{char}(k) = 0$, every k -group scheme G is geometrically reduced.*

Indeed, if $g \in G$ and K is an algebraic closure of $\kappa(g)$, one has $\mathcal{O}_{G,e} \otimes_k K \simeq \mathcal{O}_{G_K, g_K}$, so it suffices to show that $\mathcal{O}_{G,e} = \mathcal{O}_{G^0,e}$ is geometrically reduced. One is thus reduced to the case where G is connected, hence quasi-compact (2.6.5). Then the result follows from 6.5 and Cartier's theorem for algebraic groups (cf. VIB, 1.6.1 or [DG70] §II.6, Th. 1.1).

Corollary 6.10. *Let G be a quasi-compact k -group. Suppose k algebraically closed.*

- (i) *Let $f : G \rightarrow H$ be a faithfully flat morphism of k -groups. Then the induced map $G(k) \rightarrow H(k)$ is surjective.*
- (ii) *The set of rational points is dense in G .*

For the proof, we refer to [Per75], V 3.7 & 3.9.

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